Unit

33 Important organic substances

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Aspirin
- Detergents
- Polyesters
- Nylons

33.1 Introduction

Organic substances play an important role in our daily lives. They are found all around us, as they form the basis of many products that we use today. Examples of daily products made from organic substances include food, clothes, plastics, drugs, detergents, fuel and many others.

In this unit, we are going to look at the chemistry and chemical structures of some important organic substances.

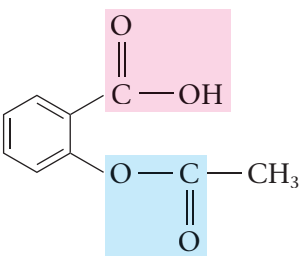
33.2 Aspirin — a common painkiller



Fig. 33.1 Aspirin is a painkiller

Have you ever had a headache? If so, you have probably swallowed some kind of painkiller. Aspirin is a common example (Fig. 33.1). It has been used for over one hundred years.

The active ingredient of aspirin tablets is acetylsalicylic acid, its structure is shown below:



DSE 2012 Paper 1A Q32

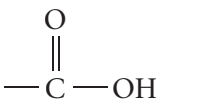
acetylsalicylic acid
or
2-ethanoyloxybenzoic acid

Reference Website

Refer to the following website for additional information about aspirin:
http://health.howstuffworks.com/medicine/medication/aspirin.htm
(accessed July 2014)

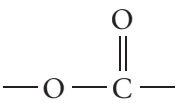
Acetylsalicylic acid contains two functional groups:

DSE 2014 Paper 1A Q32



carboxyl group N1

and



ester group

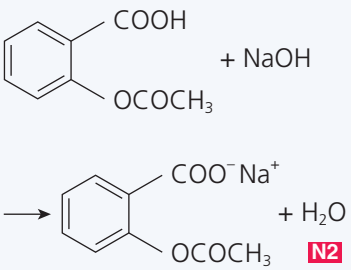
N1 Aspirin reacts with sodium carbonate solution to give carbon dioxide gas because it contains a –COOH group.

Do you know ?

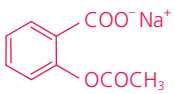
Soluble aspirin

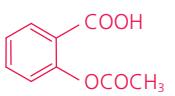
For aspirin to work, it must dissolve inside the stomach before it can get into the bloodstream. However, the aspirin discussed above is relatively non-polar. It is only slightly soluble in water.

To make aspirin work more quickly, scientists have modified aspirin by reacting it with sodium hydroxide. The sodium salt, soluble aspirin, is produced. This is soluble because it contains ions.



Soluble aspirin works faster as it dissolves very rapidly and can enter the bloodstream much more quickly than the non-soluble type.

N2 

can be converted to 

(i.e. acetylsalicylic acid) by the addition of HCl(aq).

Why take aspirin?

Apart from its use as a painkiller, there are other reasons to take small doses of aspirin.

- It lowers the body temperature quickly if you have a fever.
- It thins the blood, which reduces the risk of blood clots and heart attack.
- It reduces inflammation.
- It protects against bowel cancer.
- It slows down senile dementia (the early loss of some brain function in older age).

Problems with aspirin

Aspirin reduces the tendency for the blood to clot. Thus, an overdose can cause stomach ulcers and internal bleeding.

Aspirin should not be given to children under 16. This is because it can lead to Reye's syndrome, which is a disease that affects all organs, especially the liver and the brain, and can be fatal.

Many people now prefer to use *paracetamol* as a painkiller (Fig. 33.2). Paracetamol does not have the irritant action of aspirin on the stomach. It reduces fever but does not have the anti-inflammatory action of aspirin.



Fig. 33.2 'Panadol' is a trade name of the common painkiller paracetamol

paracetamol 撲熱息痛

33.3 Detergents

We have discussed surface tension of water in Topic 6 Microscopic World II.

Reference Website

Refer to the following website for the history of soaps and detergents:
http://inventors.about.com/library/inventors/blsoap.htm
(accessed July 2014)

Water, the liquid commonly used for cleansing, has a high **surface tension**. This tension causes water to bead up on surfaces (such as glass and fabric) (Fig. 33.3) and thus slows the wetting of the surface and inhibits the cleansing process. In the cleansing process, surface tension must be reduced so that water can spread and wet surfaces. That is why we need **detergents**. A detergent is a substance which improves the cleansing properties of water.



Fig. 33.3 Surface tension causes water to bead up on the surface of a piece of clothing

There are two types of detergent:

CE 2008 Paper 2 Q35
CE 2010 Paper 2 Q41
CE 2011 Paper 2 Q47

1 **Soapy detergents** They are commonly called 'soaps' (Fig. 33.4). They are made from natural fats and oils.

DSE 2014 Paper 1A Q34

2 **Soapless detergents or synthetic detergents** They are made from chemicals derived from petroleum (Fig. 33.5).



Fig. 33.4 Bars of soap



Fig. 33.5 Soapless detergents

surface tension 表面張力　　detergent 清潔劑　　soapy detergent 皂性清潔劑　　soapless detergent 非皂性清潔劑
synthetic detergent 合成清潔劑

33.4 How do detergents help water to clean?

33.1
Investigating the properties of detergents.

Video
Surface tension of water and using soap or detergent as a wetting agent

N3 Students should be able to identify the hydrophobic and hydrophilic parts of a detergent.

A detergent helps water to remove dirt by

- its ability to act as a **wetting agent**; and
- its **emulsifying action**.

Fig. 33.6 gives a diagrammatic representation of a typical anionic detergent. A typical anionic detergent has a hydrocarbon 'tail' which is soluble in grease and oil, and an anionic 'head' which is soluble in water. The hydrocarbon 'tail' is said to be **hydrophobic** and the anionic 'head' is said to be **hydrophilic**. N3

CE 2007 Paper 1 Q13
CE 2009 Paper 2 Q45

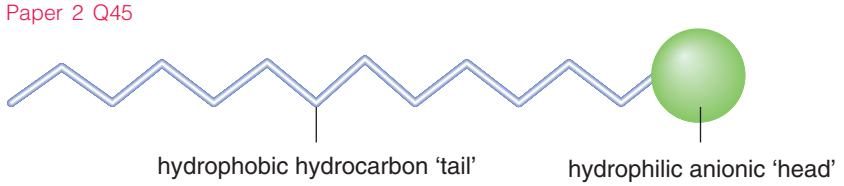


Fig. 33.6 Structure of a typical anionic detergent

CE 2007 Paper 1 Q13

Fig. 33.7 shows the structure of a typical soap particle. The hydrophilic anionic 'head' is a $-\overset{\overset{O}{\|}}{C}-O^-$ group.

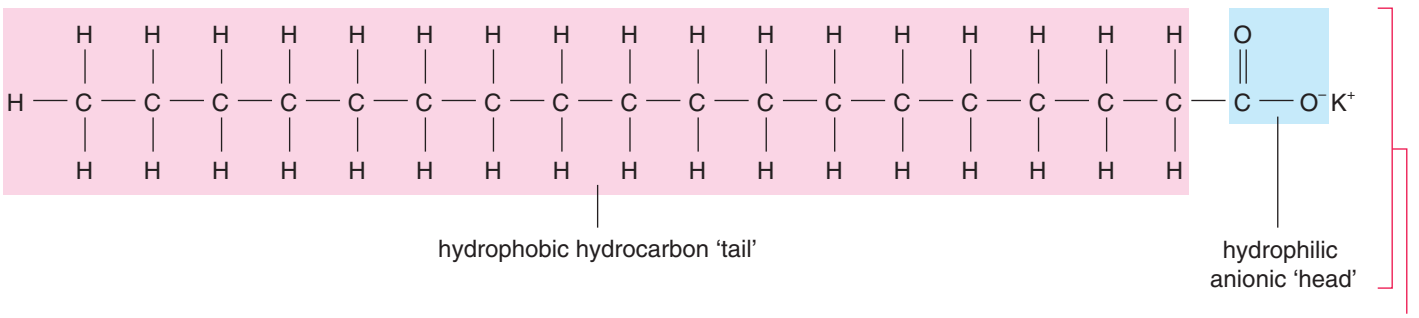
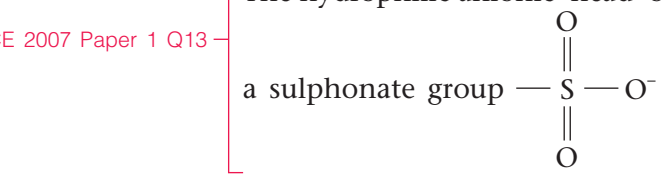
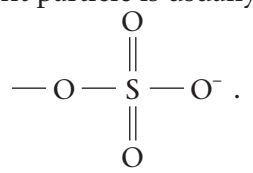


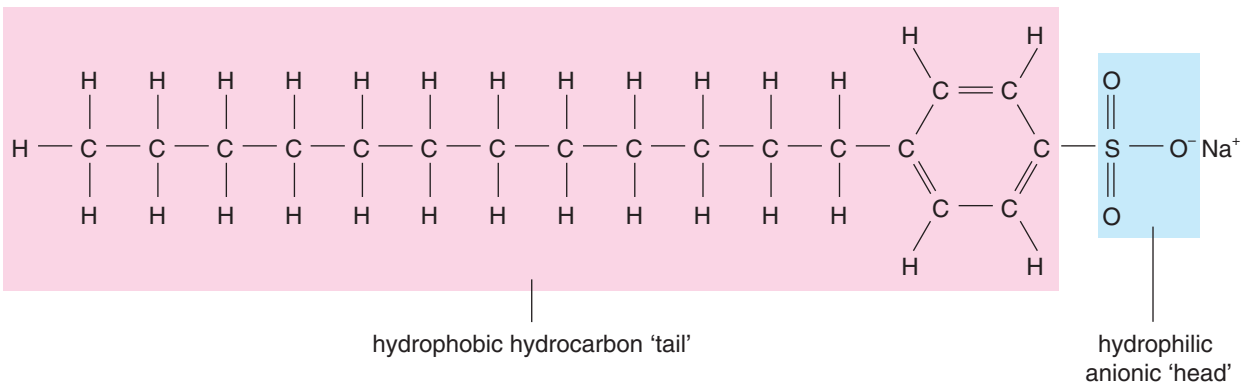
Fig. 33.7 Structure of a typical soap particle

CE 2005 Paper 2 Q42
CE 2008 Paper 2 Q35

wetting agent 濕潤劑 emulsifying action 乳化作用 hydrophobic 疏水性 hydrophilic 親水性

CE 2007 Paper 1 Q13

Fig. 33.8 shows the structures of two common soapless detergent particles. Their structures are quite similar to that of a soap particle. The hydrophilic anionic 'head' of a soapless detergent particle is usually a sulphonate group  or sulphate group  .



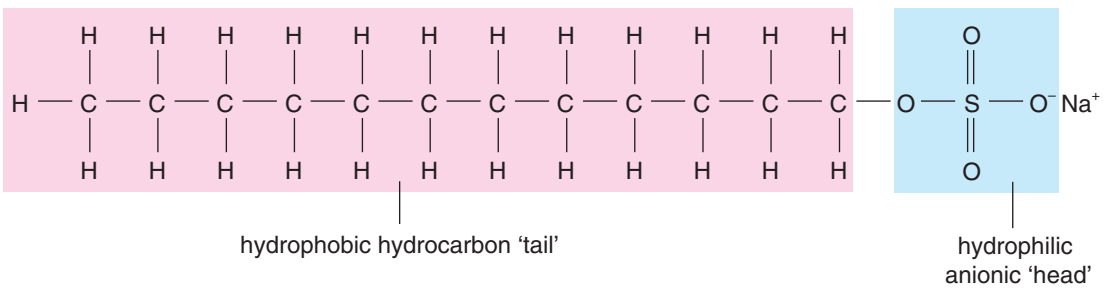


Fig. 33.8 Structures of two common soapless detergent particles

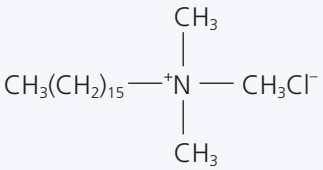
Do you know

Different types of synthetic detergents

Anionic synthetic detergents are the most common type of detergents. There are also other types of synthetic detergents such as cationic detergents and non-ionic detergents.

A cationic detergent has a positively charged 'head'.

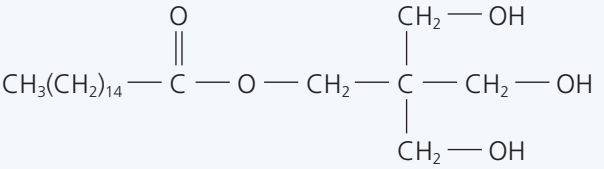
e.g.



The hydrocarbon 'tail' of this detergent is hydrophobic while the charged N atom is hydrophilic.

A non-ionic detergent also has a hydrocarbon 'tail', but the head is a polar segment.

e.g.



The polar segment of this detergent consists of three –OH groups and one  group.

33.5 The wetting and emulsifying properties of detergents in relation to their structures

The wetting property of detergents

When a drop of water is dropped onto a piece of fabric, the forces between water molecules hold them into a drop (Fig. 33.9).

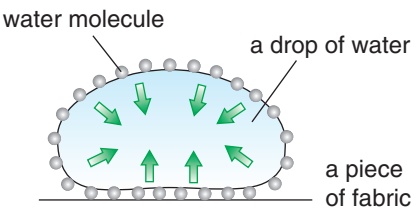


Fig. 33.9 Forces between water molecules hold them into a drop

The situation is different when a drop of detergent solution is dropped onto a piece of fabric. A large number of the detergent particles tend to stay at the surface, with their hydrophobic hydrocarbon ends sticking out. The intervening detergent particles separate the water molecules, decreasing the attractive forces between them. The water no longer beads up but spreads, and thus wets the surface more thoroughly (Fig. 33.10).

CE 2006 Paper 2 Q26
CE 2007 Paper 1 Q13
CE 2008 Paper 2 Q35

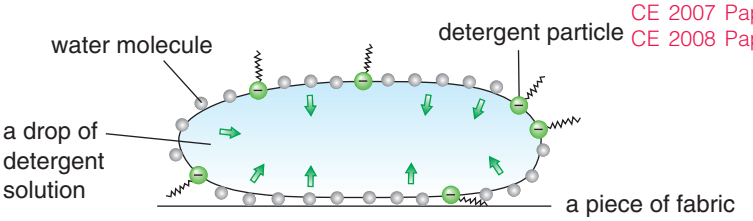


Fig. 33.10 The detergent acts as a wetting agent

CE 2006 Paper 2 Q26
CE 2007 Paper 1 Q13
CE 2007 Paper 2 Q50
CE 2009 Paper 1 Q12(d)
CE 2009 Paper 2 Q45
DSE 2014 Paper 1A Q34

The emulsifying property of detergents

N4 Refer to page T58.

In an oil-water mixture, detergent particles take the positions as shown in Fig. 33.11a (before shaking). The hydrocarbon 'tails' are soluble in the oil and the anionic 'heads' are soluble in the water.

When we shake the mixture, oil droplets form. Each droplet is surrounded by detergent particles with the anionic 'heads' in the water (Fig. 33.11b). The detergent particles prevent the oil droplets from coming together again. Thus, the oil droplets remain suspended in the water. An **emulsion** forms. N4

The anionic heads have like charges. They repel each other and prevent the oil droplets from coming together again.

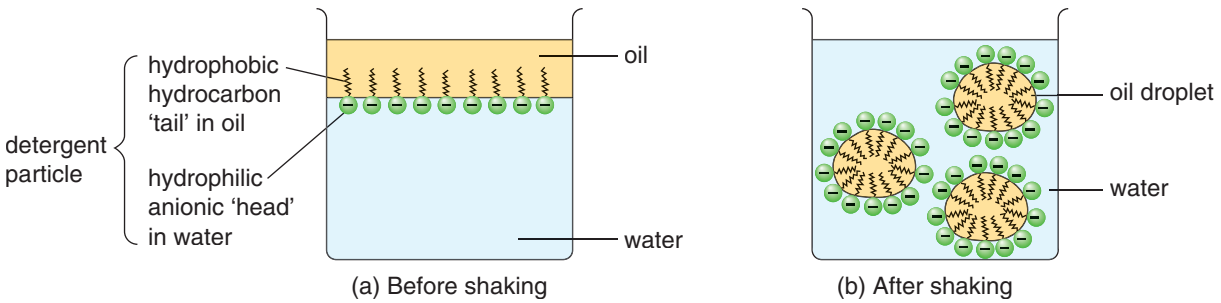


Fig. 33.11 How detergent particles arrange themselves in an oil-water mixture

33.6 The cleansing action of detergents

Animation
The emulsifying process of a detergent

Reference Website
Refer to the following website for an animation illustrating the cleansing action of a detergent:
http://resources.hkedcity.net/resource_detail.php?rid=1171602452
(accessed July 2014)

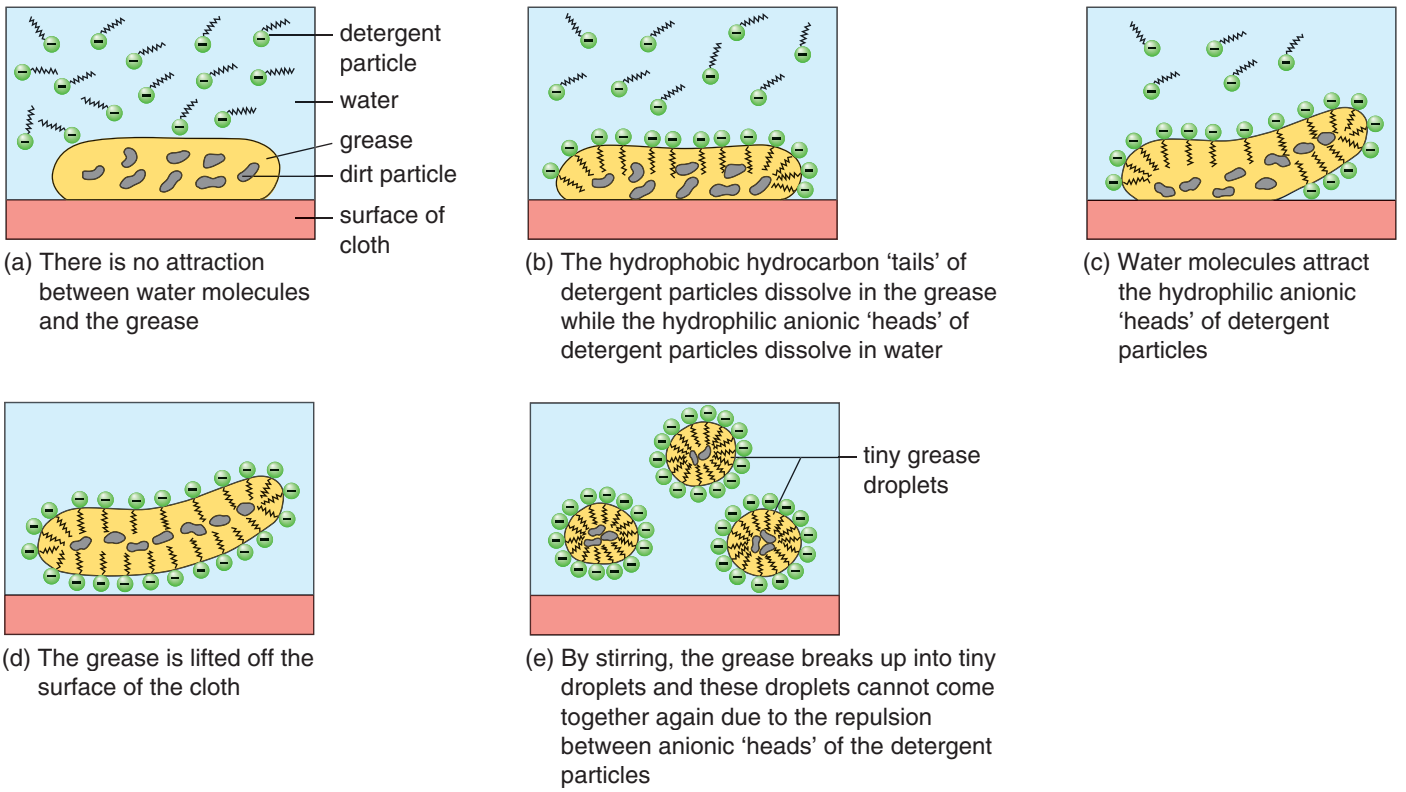
Grease binds the dirt particles on a surface. Grease is insoluble in water. Therefore water alone will not remove the grease and dirt. How does a detergent clean?

Imagine that there is a greasy mark on a piece of cloth (Fig. 33.12a). When we add a detergent solution to the cloth, the hydrocarbon 'tails' of the detergent particles dissolve in the grease. The anionic 'heads' of the detergent particles are insoluble in the grease and remain outside (Fig. 33.12b). The surrounding water molecules attract the anionic 'heads' and lift the grease off the surface (Fig. 33.12c–d).

By stirring, the grease (which carries the dirt particles) breaks up into tiny droplets suspended in the water. These tiny droplets cannot come together again due to the *repulsion* between the anionic 'heads' of detergent particles on their surfaces. An emulsion forms (Fig. 33.12e). Rinsing washes away the greasy suspension, and leaves the surface clean.

DSE 2012 Paper 1B Q14(b)

emulsion 乳狀物　　repulsion 排斥



(a) There is no attraction between water molecules and the grease

(b) The hydrophobic hydrocarbon 'tails' of detergent particles dissolve in the grease while the hydrophilic anionic 'heads' of detergent particles dissolve in water

(c) Water molecules attract the hydrophilic anionic 'heads' of detergent particles

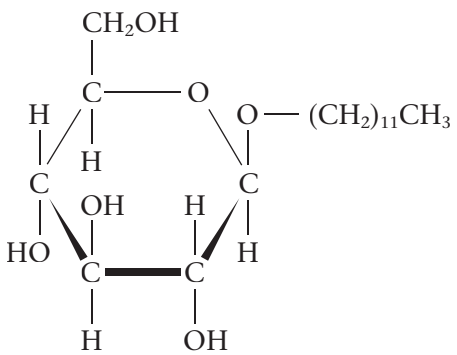
(d) The grease is lifted off the surface of the cloth

(e) By stirring, the grease breaks up into tiny droplets and these droplets cannot come together again due to the repulsion between anionic 'heads' of the detergent particles

Fig. 33.12 The cleansing action of detergents

Example 33.1

Q A new detergent particle has the structure shown below:



Explain why this detergent can be used to clean a greasy mark on a piece of cloth.

A The detergent particle has a hydrocarbon chain which is soluble in grease and a ring which is soluble in water (due to hydrogen bonds between the ring and water molecules). Thus, the detergent particle can help lift the grease off the cloth.

c) Before shaking an oil-water mixture, the hydrocarbon 'tails' of detergent particles are soluble in the oil and the anionic 'heads' are soluble in the water.
Upon shaking, oil droplets form. Each droplet is surrounded by detergent particles with the anionic 'heads' in the water.
Repulsion between the anionic heads prevents the oil droplets from coming together again. Thus, the oil droplets remain suspended in the water. An emulsion is formed.

Practice 33.1

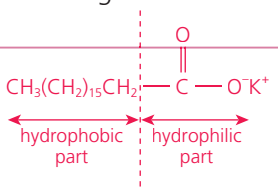
The structures of two detergent particles are shown below:

Detergent I $CH_3(CH_2)_{10}CH_2-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-O^-K^+$

Detergent II $CH_3(CH_2)_{15}CH_2-\overset{\overset{O}{\|}}{C}-O^-K^+$

a) Which of the detergents is / are soapless detergent(s)? Detergent I

b) Identify the hydrophilic part and hydrophobic part of detergent II.



c) With reference to the structure of detergent II, explain why an emulsion is formed when the detergent is shaken with an oil-water mixture.

d) In acidic environments, detergent II loses its cleansing function because it forms an insoluble organic acid.

i) Write the structural formula of the organic acid formed. $CH_3(CH_2)_{16}COOH$

ii) With the help of an ionic equation, explain why sodium carbonate can improve the cleansing abilities of detergent II in acidic environments.
Sodium carbonate can reduce the acidity in the acidic environment.
$2H^+(aq) + CO_3^{2-}(aq) \longrightarrow CO_2(g) + H_2O(l)$

CE 2011 Paper 1 Q11(b)

33.7 Making soaps and soapless detergents

Making soaps

CE 2005 Paper 2 Q32
CE 2006 Paper 2 Q42
CE 2007 Paper 1 Q13
CE 2009 Paper 1 Q12(a)

Soaps (Fig. 33.13) are made by a reaction called **saponification**. This reaction involves boiling fats or oils with a strong alkali, usually sodium hydroxide solution or potassium hydroxide solution.



Fig. 33.13 A collection of different kinds of soap — the different colours are obtained by adding dyes to the soaps

saponification 皂化作用

Video
Preparing a soap and testing its properties (part A)

CE 2005 Paper 2 Q32
CE 2009 Paper 1 Q12(b)
CE 2011 Paper 2 Q39

33.2
Preparing a soap and testing its properties.

We can make a 'soap cake' in the laboratory using vegetable oil and concentrated sodium hydroxide solution (Fig. 33.14). First gently heat, while stirring, a mixture of castor oil and concentrated sodium hydroxide solution for about 20 minutes. Most of the soap formed dissolves in the reaction mixture. Then add concentrated sodium chloride solution to the mixture. This lowers the solubility of soap in water. The soap separates from the solution and floats on the surface. N5

This process is called the salting-out of soap.

Obtain the soap by filtration. Then wash the soap with a little distilled water and allow to dry.

N5 Examination questions often ask about the purpose of adding concentrated sodium chloride solution in soap preparation.

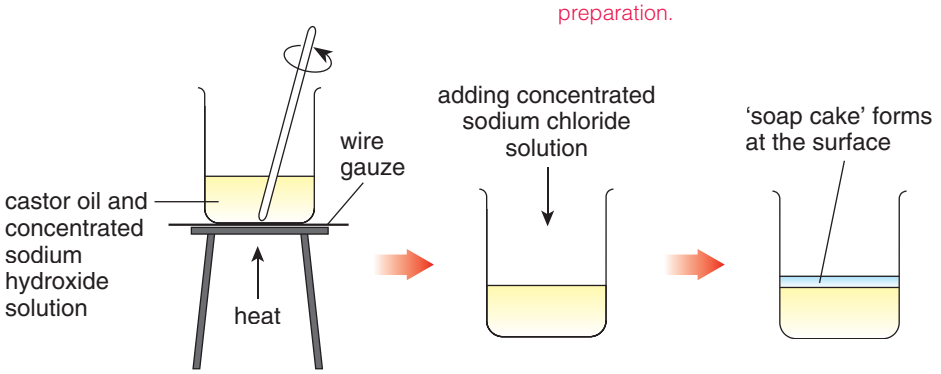


Fig. 33.14 Making a soap in the laboratory

Do you know

How are soaps made commercially?

To make soaps commercially, various vegetable oils or chopped up fats (residues from *butcher shops*, *abattoirs* and other food processing factories) are heated with about 30% sodium hydroxide solution until all the oil or fat has dissolved. Then the solution is cooled and mixed with nearly saturated brine (NaCl solution). Upon standing this mixture settles into two layers — curds of soap on top of a solution.

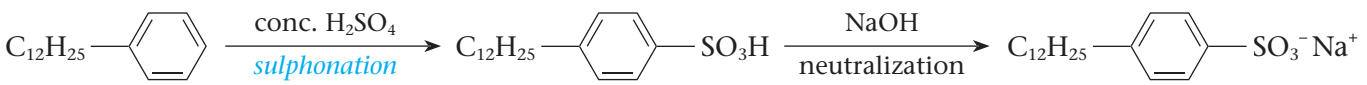
After the curds of soap are removed and washed, various additives such as perfumes and colouring agents are introduced. Then the mixture is pressed and cut or moulded into bars for packaging and use.

Making soapless detergents

CE 2007 Paper 1 Q13

Soapless detergents are made from hydrocarbons obtained from petroleum. The hydrocarbons are first reacted with very concentrated sulphuric acid and the product is then neutralized with sodium hydroxide.

For example,



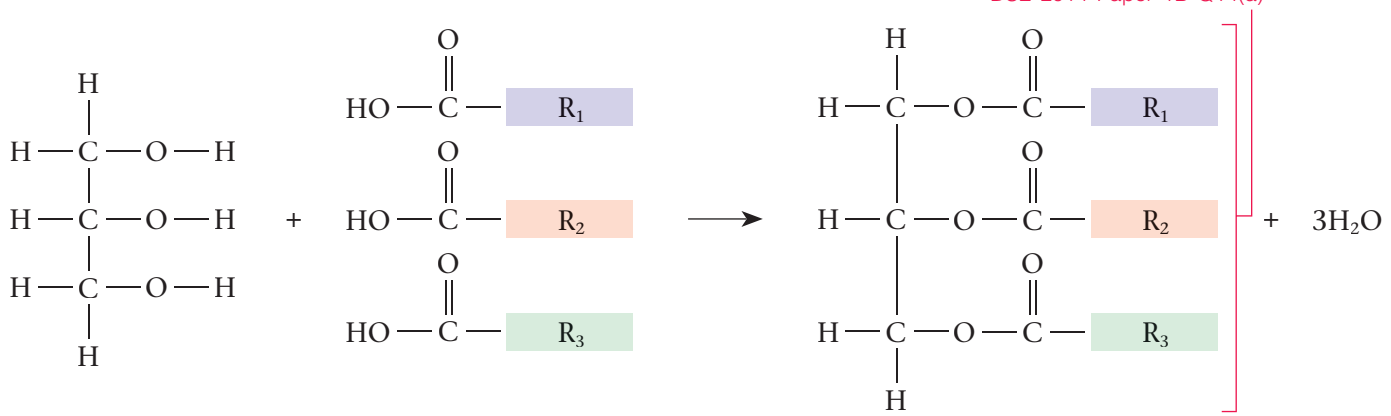
hydrocarbon obtained from petroleum

butcher shop 肉店 abattoir 屠宰場 sulphonation 磺化作用

33.8 Fats and oils

Fats and oils are triesters of long-chain carboxylic acids (fatty acids) and the alcohol propane-1,2,3-triol (commonly known as glycerol). There are three –OH groups in a glycerol molecule. Three carboxylic acid molecules can join to the –OH groups via ester linkages, forming a triester molecule.

DSE 2014 Paper 1B Q14(a)



N6 Refer to page T58.

A triester formed from glycerol and three long-chain carboxylic acids is called a **triglyceride**. Fats and oils are mixtures of different triglycerides.

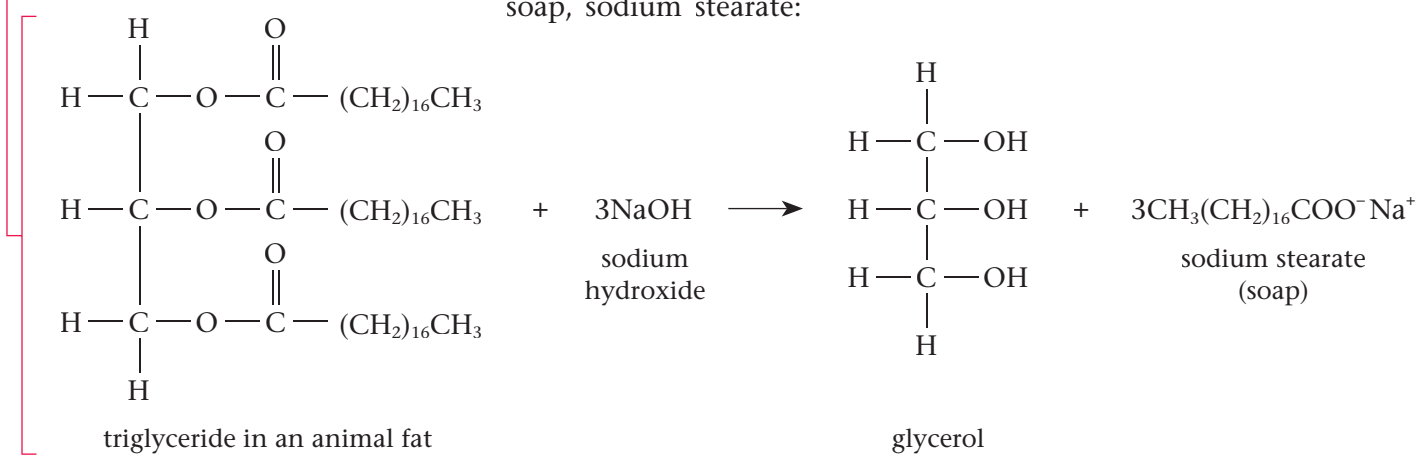
Making soaps from fats and oils

Refer to the hydrolysis of esters discussed in Unit 31.

To make soaps, fats or oils are boiled with a strong alkali. Alkaline hydrolysis of triglycerides (i.e. saponification) produces glycerol and a mixture of salts of long-chain carboxylic acids. These salts of long-chain carboxylic acids are soaps.

CE 2009 Paper 1 Q12(c)
ASL 2008 Paper 1 Q6(a)

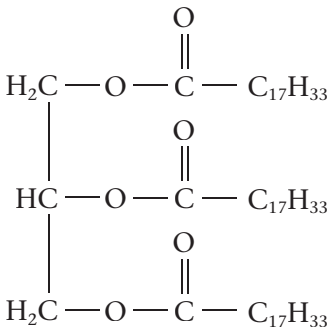
The following equation represents the production of a common soap, sodium stearate:



triglyceride 甘油三酯

Example 33.2

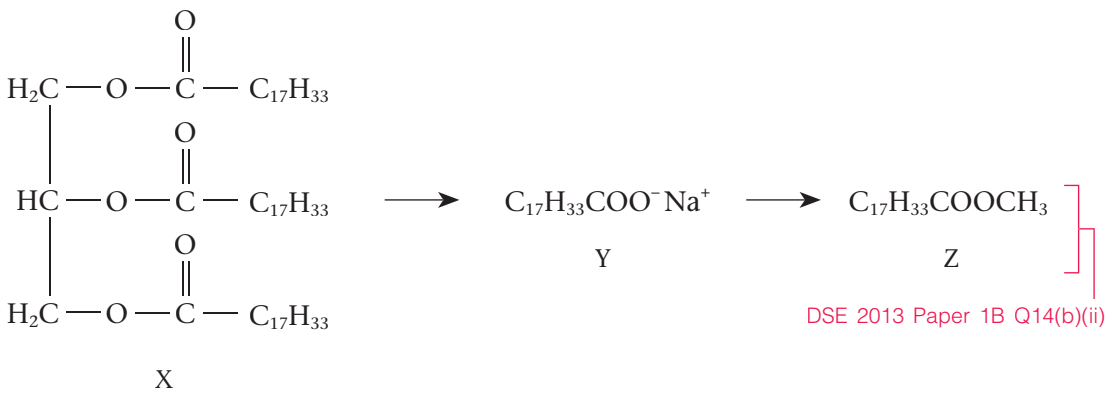
Q The structure of one component X of a vegetable oil is shown below:



All alkyl groups in X are straight chains.

a) Name all the functional groups in X. ASL 2005 Q12(a)(II)(i)

b) Study the following conversions starting from X.



DSE 2013 Paper 1B Q14(b)(ii)

i) Suggest how X can be converted to Y.

ii) Suggest ONE use of Y.

DSE 2013 Paper 1B Q14(c)

iii) With reference to their relative molecular masses and properties, explain why Z can be used as a fuel for cars, but X cannot.

Continued on next page

A a) Ester functional group and carbon-carbon double bond

b) i) Boiling with a strong alkali

ii) As a soap

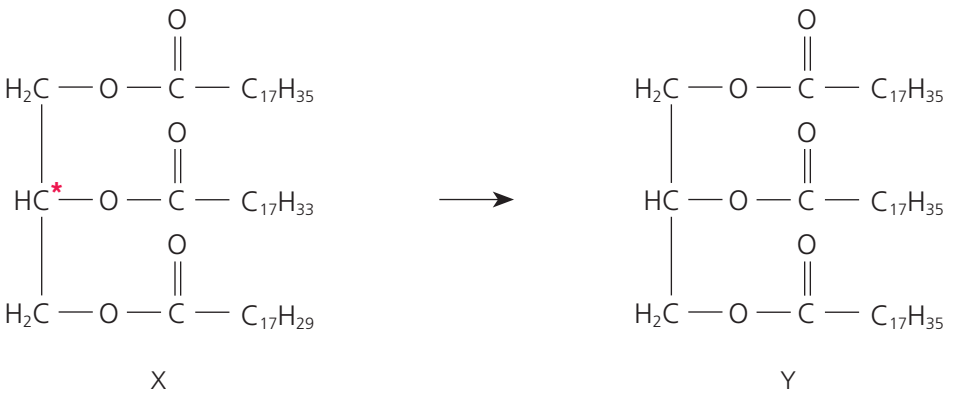
iii) The relative molecular mass of Z is lower than that of X. The molecular size of Z is smaller than that of X. The van der Waals' forces between molecules of Z are weaker than those between molecules of X. Thus, Z can be vaporized more easily.

The side-chains of X can entangle together and the relative motion between molecules will be hindered. On the other hand, Z consists of straight chain molecules and the molecules can slide past one another more easily. Thus, Z is less viscous than X.

Finally, the more volatile Z burns more completely than X.

Practice 33.2

The diagram below shows the conversion of an oil molecule X to a fat molecule Y.



All alkyl groups in X are straight chains.

a) i) X contains a chiral carbon. Mark the chiral carbon. ⌐ DSE 2012 Paper 1B Q14(a)(i)
The chiral carbon is marked with *.

ii) State the reagents needed for converting X to Y. ⌐ DSE 2013 Paper 1B Q14(a)
$H_2(g)$ / Pt catalyst

iii) The conversion involves a change in optical activity. Explain. ⌐ DSE 2012 Paper 1B Q14(a)(ii)
a) iii) X has one chiral carbon and hence it is optically active.
Y does not have any chiral carbons and hence it is optically inactive.
Thus, the conversion involve a change in optical activity.

b) Y can be used to make soap.

i) Write the chemical equation involved for the formation of soap from Y. N7 ⌐ DSE 2013 Paper 1B Q14(b)(i)
Refer to page T58.

ii) Explain how the soap produced helps water to clean.
A detergent helps water to remove dirt by
- its ability to act as a wetting agent; and
- its emulsifying action.

N7 Refer to page T59.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Environmental issues related to the use of detergents

N8 Refer to page T59.

The discharge of wastewater containing detergents into drains has environmental consequences.

Problems with non-biodegradable detergents

When soapless detergents were first made, they were derived from branched-chain hydrocarbons (Fig. 33.15). The branched-chain detergents were **non-biodegradable** and their residues remained in waterways and wastewater treatment plants. By the 1950s, drains and rivers often carried persistent mounds of *foam*, and the water became toxic to small organisms living in the water.

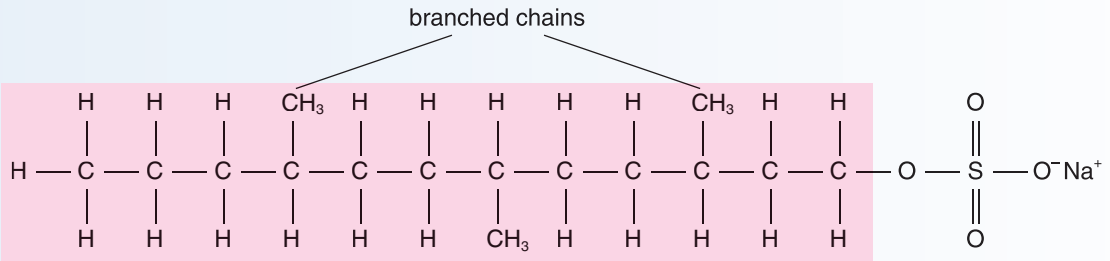


Fig. 33.15 The structure of a non-biodegradable soapless detergent (with branched hydrocarbon chains)

Problems with biodegradable detergents

Most soapless detergents nowadays are **biodegradable**. These detergents contain straight (unbranched) hydrocarbon chains. Being biodegradable, they can decompose naturally and readily by bacterial action. However, they produce some other problems.

CE 2009 Paper 2 Q45
CE 2011 Paper 2 Q47

1 When too much biodegradable detergent is added to water, the microorganisms multiply themselves and use up the dissolved oxygen in water. This is called oxygen *depletion*. As oxygen in the water has all been used up, some of the harmless microorganisms die and leave only those microorganisms that produce toxins harmful to people, animals and the environment.

CE 2009 Paper 2 Q50
CE 2011 Paper 1 Q11(a)(ii)

2 Many detergents have phosphate '*builders*' (Fig. 33.16) added to increase cleansing power. However, increased phosphate concentration in natural waterways can lead to harmful algal bloom.

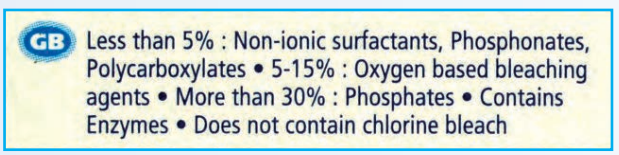


Fig. 33.16 The label of a detergent with phosphates added

'Builders' soften the water and prevent dirt from going back into the material again.

Video

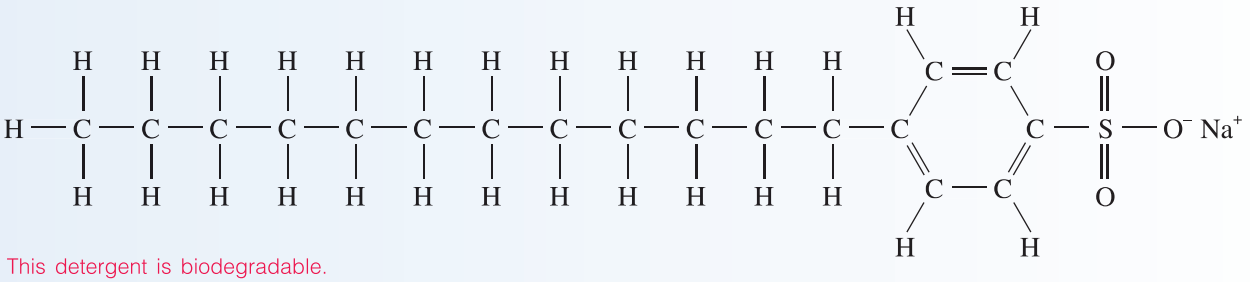
A greener alternative to biodegradable detergents

In many countries, there are restrictions on the amount of phosphates allowed in washing powders and liquids.

non-biodegradable 生物不可降解的　foam 泡沫　biodegradable 生物可降解的　depletion 消耗　builder 助潔劑

Questions

1 The structure of a commonly-used detergent is shown below.



This detergent is biodegradable.
Producing this detergent can save food.

Suggest TWO advantages of this detergent.

2 Suppose algae grow so fast that they form a thick layer on the surface of a pond (Fig. 33.17).

Suggest how this would affect the living organisms in the pond.

Algae cut off the light supply. Plants underneath the water surface cannot carry out photosynthesis and die. They use up dissolved oxygen upon decomposition and cause smells and plagues.

Deprived of oxygen, shelter and food, the fish larvae starve.

CE 2011 Paper 1 Q11(c)



Fig. 33.17 A pond covered with algae

33.9 Polyesters

Polyesters are polymers in which the monomer units are linked by ester linkages (—C(=O)—O—).

When a carboxylic acid with one –COOH group reacts with an alcohol with one –OH group, an ester and water are formed and no further esterification reactions can take place. If the acid has two –COOH groups and the alcohol has two –OH groups, then the ester that is formed would still contain a –COOH group at one end and an –OH group at the other (Fig. 33.18). Further esterification reactions are possible. A polyester can form.

polyester 聚酯

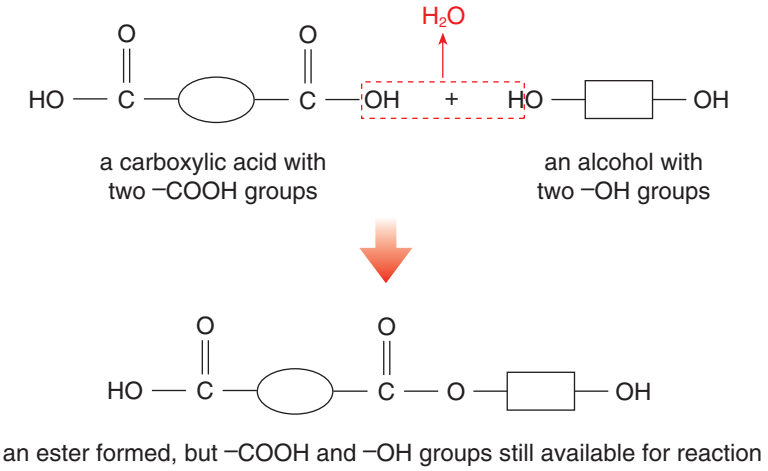


Fig. 33.18 Esterification between a carboxylic acid with two –COOH groups and an alcohol with two –OH groups

Poly(ethylene terephthalate) (PET)

N9 Examination questions often ask about structures of the monomers for producing PET.

Poly(ethylene terephthalate) (PET) is one of the most important polyesters. The names and structural formulae of the monomers for producing this polyester are shown below.

CE 2010 Paper 1 Q12(a)

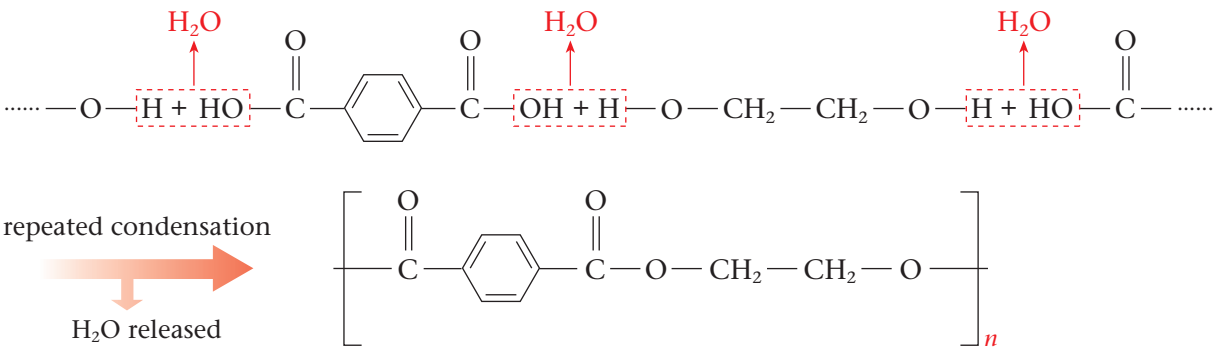


terephthalic acid
or
benzene-1,4-dicarboxylic acid

ethylene glycol
or
ethane-1,2-diol N9

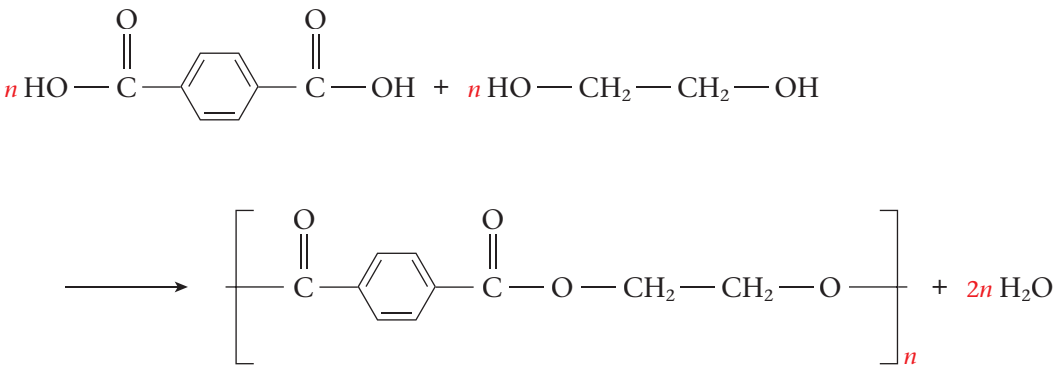
CE 2005 Paper 2 Q49
CE 2011 Paper 2 Q34

During the production of this polyester, repeated condensation reactions occur between the –COOH groups of the dicarboxylic acid molecules and the –OH groups of the diol molecules. Water molecules are released.



poly(ethylene terephthalate) (PET) 聚對苯二甲酸乙二酯

We can represent the above reaction by the following equation:



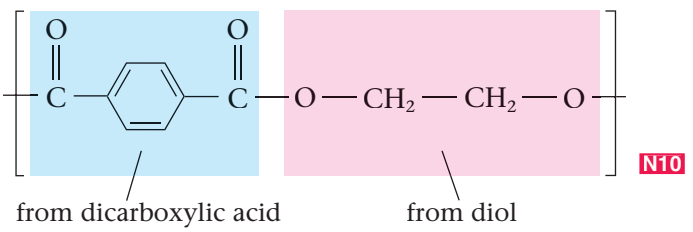
CE 2010 Paper 1 Q12(c)

The above reaction is a special type of condensation reaction — **condensation polymerization** — in which a polymer is formed.

We will discuss condensation polymerization further in Topic 14 Materials Chemistry.

Polymers that are made by condensation polymerization are called condensation polymers. Thus, poly(ethylene terephthalate) is a condensation polymer.

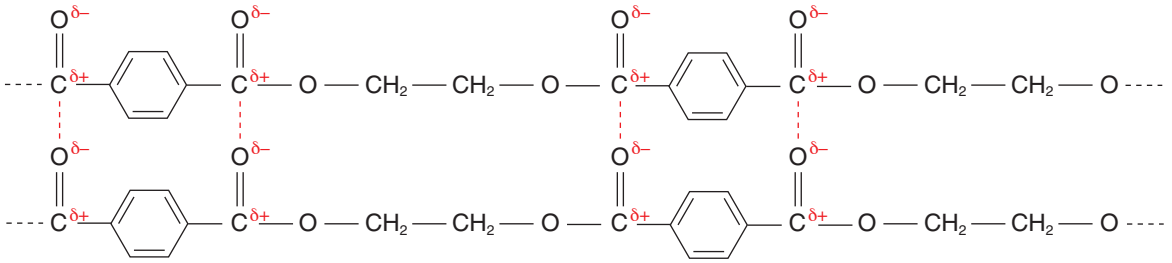
The repeating unit of poly(ethylene terephthalate) is



N10 The repeating unit of a polymer
Examination questions often give the repeating unit of a polymer and ask about information of the polymer. For example,
- whether it is an addition polymer or a condensation polymer;
- whether it is formed from one monomer or two different monomers;
- whether the polymer has a giant covalent structure.

Poly(ethylene terephthalate) is a thermoplastic. It consists of separate long polymer chains. The C=O groups in the polymer chains are polar and there is considerable intermolecular attraction between the polymer chains (Fig. 33.19). Hence the polymer has a great strength.

condensation polymerization 縮合聚合作用

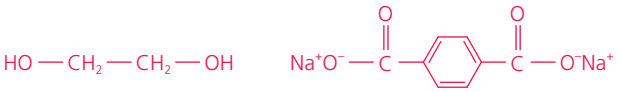


key:

---- intermolecular attraction

Fig. 33.19 Intermolecular attraction between PET polymer chains

N11 Alkaline hydrolysis of PET gives the following products:



CE 2010 Paper 1 Q12(e)(i)
ASL 2010 Paper 2 Q3(b)
AL 2010 Paper 2 Q4(d)(ii)

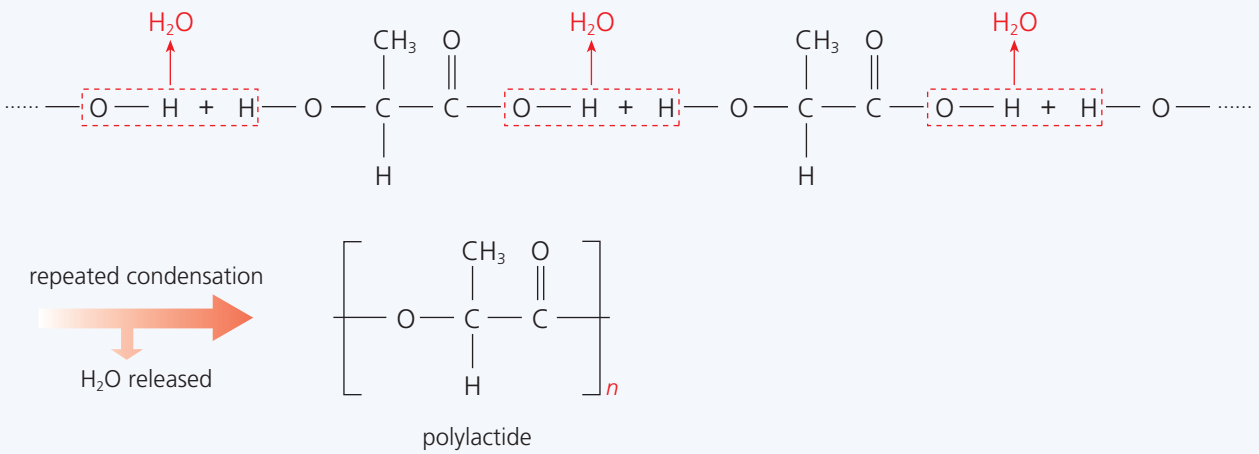
Polyesters are hydrolyzed readily by alkalis. If we spill some dilute alkali on a fabric made from polyester, the ester linkages would break. A hole would appear. Polyesters are readily degraded in the environment because they undergo hydrolysis in the presence of an acid or alkali to give the monomers. N11

Do you know

Polyesters from one type of monomer

Polyesters can also be made from just one type of monomer containing –COOH and –OH groups.

For example, *polylactide* is made from the monomer lactic acid (HO—C(CH$_3$)(H)—COOH).



polylactide

Polylactide is biodegradable and there is much interest in its use for food and drink containers.

polylactide 聚乳酸

Uses of poly(ethylene terephthalate)

The synthetic fibres, *Terylene* and *Dacron*, are formed from the polyester by spinning. For clothing and bedding, the polyester is often blended with cotton (Fig. 33.20). The polyester gives the material strength and *crease* resistance while the cotton allows the material to absorb some perspiration from the user.

The name Terylene is derived from the names of the two monomers: terephthalic acid and ethylene glycol.



Fig. 33.20 Clothes made of Terylene

The polyester has a great strength and good chemical resistance. It also possesses good gas and moisture barrier properties. Hence the polyester is widely used for bottling carbonated drinks (Fig. 33.21).

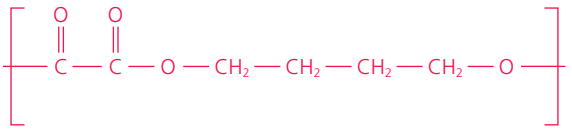


Fig. 33.21 PET bottles for carbonated drinks

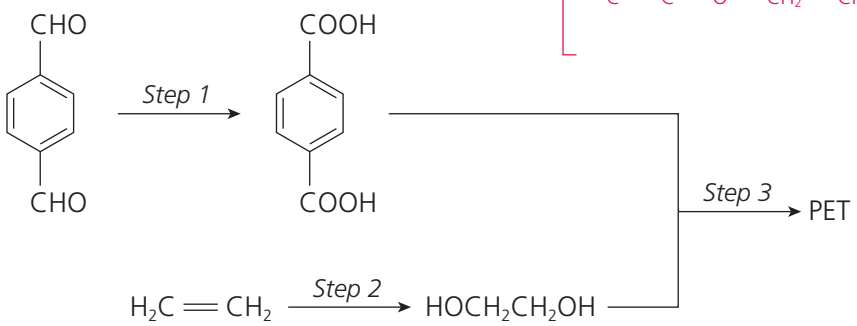
Terylene 特麗綸　Dacron 滌綸　crease 摺痕

Practice 33.3

1 Draw the repeating unit of the polymer that can be made from ethanedioic acid and butane-1,4-diol.

1 

2 Poly(ethylene terephthalate) (PET) is a polymer used in making clothing fibres and soft drink bottles. The flow diagram below shows a route for the synthesis of PET.



a) Give the reagents and conditions for each of *Steps 1*, *2* and *3*.

Step 1 — $K_2Cr_2O_7$ / H_3O^+
Step 2 — cold dilute $KMnO_4$ / OH^-
Step 3 — conc. H_2SO_4, heat

b) Explain why PET is readily degraded in the environment.
PET is a polyester. It undergoes hydrolysis in the presence of acid / alkali.

33.10 Nylon

Polyamides are polymers in which the monomer units are linked by amide linkages (). As with polyesters, polyamides are condensation polymers that can be made from two different monomers — one monomer being a diamine with two $-NH_2$ groups and the other being a dicarboxylic acid with two –COOH groups.

nylon 尼龍　　polyamide 聚酰胺

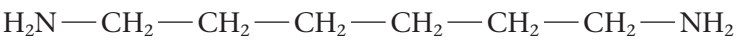
Nylon-6,6

33.3
Preparing nylon.

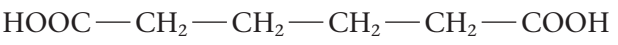
Video
Preparing nylon

The first synthetic and commercially important polyamides were various forms of nylon. A common type of nylon is called nylon-6,6 because it is made from two monomers, each of which contains six carbon atoms in the molecules. The names and structural formulae of the monomers are shown below.

CE 2006 Paper 2 Q43



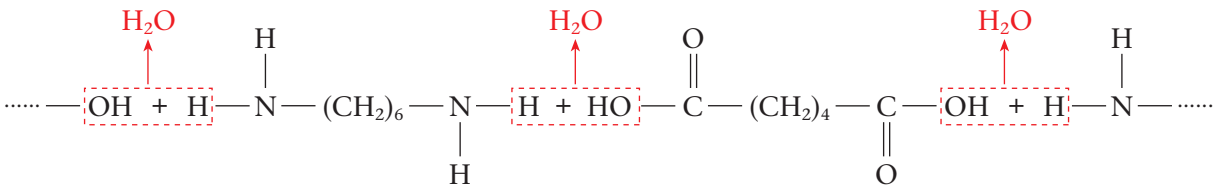
hexane-1,6-diamine

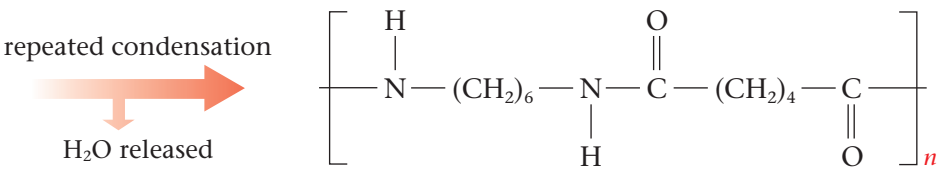


hexanedioic acid

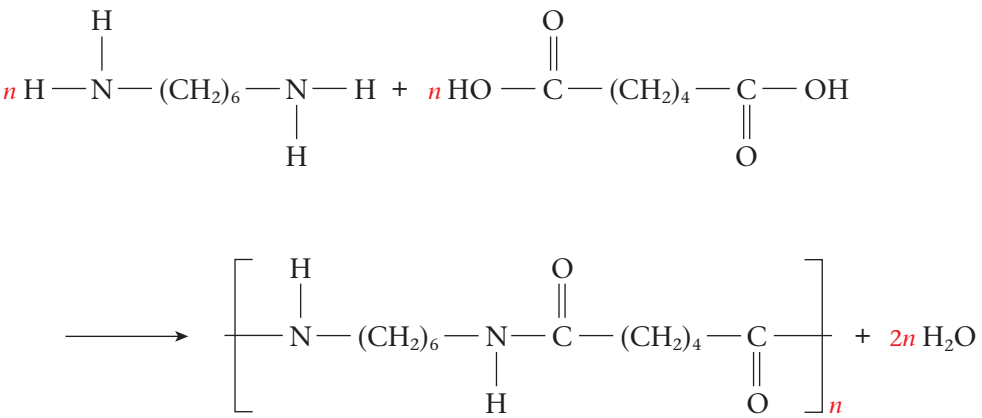
Reference Website
Refer to the following website for an animation showing the preparation of nylon from its monomers:
http://www.lstlcw.edu.hk/t9544/animation/con_polymerization/tc_con_poly.html
(accessed July 2014)

During the production of nylon-6,6, repeated condensation reactions occur between the $-NH_2$ groups of the hexane-6,6-diamine molecules and the $-COOH$ groups of the hexanedioic acid molecules. Water molecules are released.





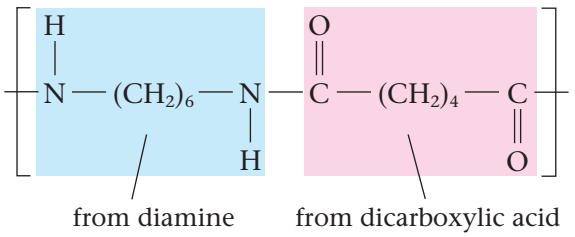
We can represent the above reaction by the following equation:



CE 2006 Paper 2 Q43
CE 2011 Paper 2 Q48

Nylon-6,6 is made by condensation polymerization. Thus, it is a condensation polymer.

The repeating unit of nylon-6,6 is



Nylon-6,6 is a thermoplastic. It consists of separate long polymer chains. Hydrogen bonds exist between its polymer chains (Fig. 33.22). This explains its high melting point and great strength.

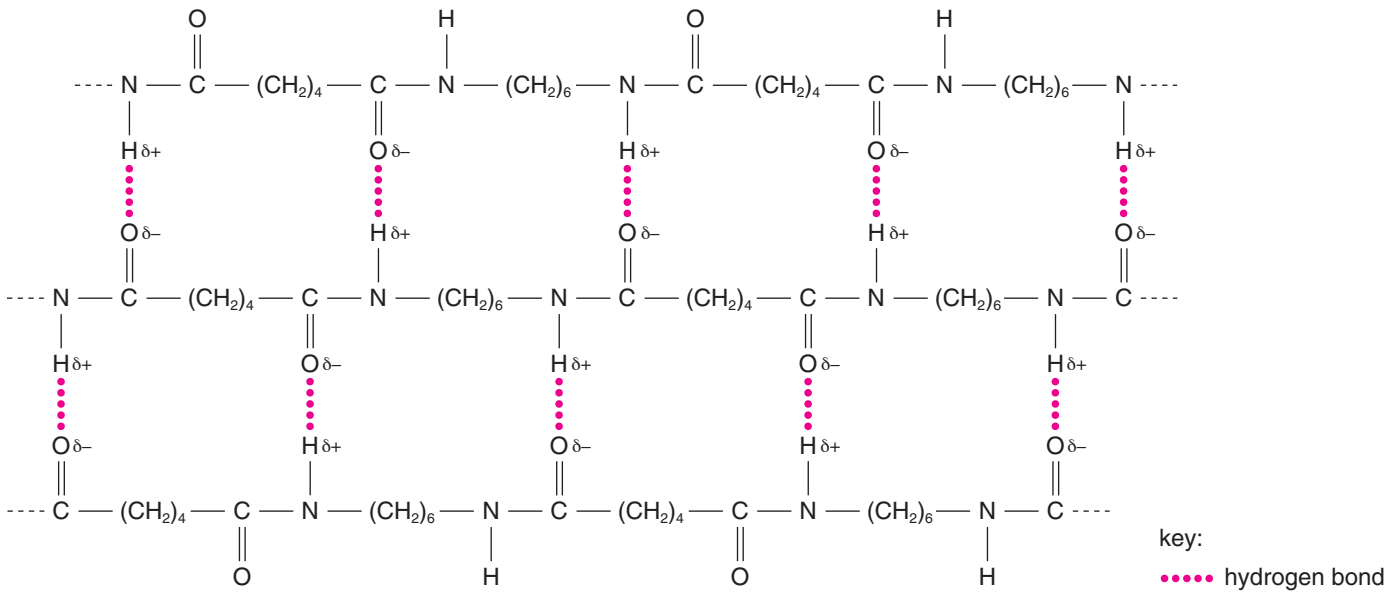


Fig. 33.22 Hydrogen bonding between the polymer chains of nylon-6,6

Polyamides are hydrolyzed readily by strong acids. If we spill some dilute sulphuric acid on a fabric made from nylon, the amide linkages would break. A hole would appear on the fabric.

Uses of nylons

Nylons have high melting points and a great strength. Nylon fibres are extensively used for clothes, ropes, fishing lines and nets, etc (Figs. 33.23–33.24). Nylon was the first man-made fabric to function as a synthetic substitute for silk.



Fig. 33.23 Nylon sportswear

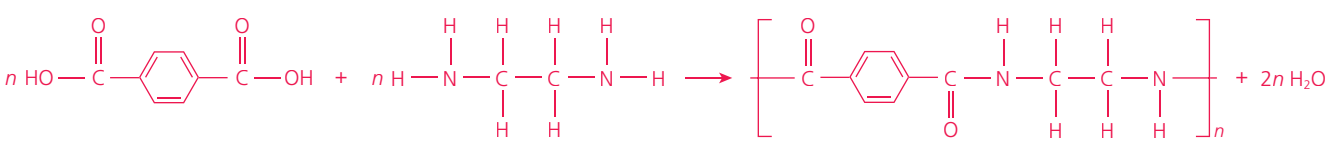


Fig. 33.24 Mountaineers' rope made from nylon

Nylons are also used for mechanical parts such as *gears* and *bearings*. These parts last for a long time because nylons are hard and strong.

Practice 33.4

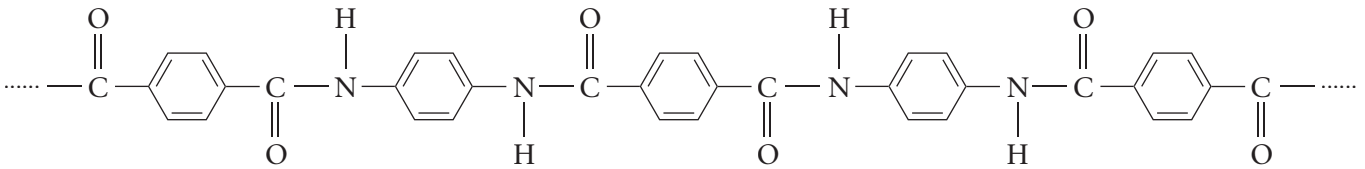
2 b)



gear 齒輪　　bearing 軸承

Example 33.3

Q Twaron is a heat-resistant, high-strength fibre used in protective clothing. A short section of the structure of Twaron is shown below.

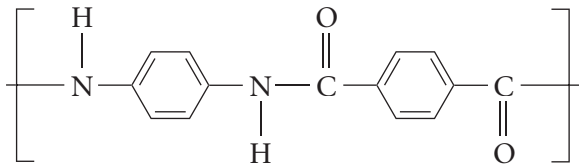


a) Draw the repeating unit of Twaron.

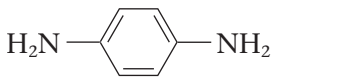
b) Draw the structures of the two monomers that could be used to prepare Twaron. N12 Refer to page T60.

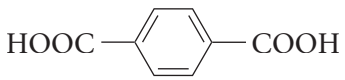
c) Explain why Twaron has a great strength.

A a)



b)



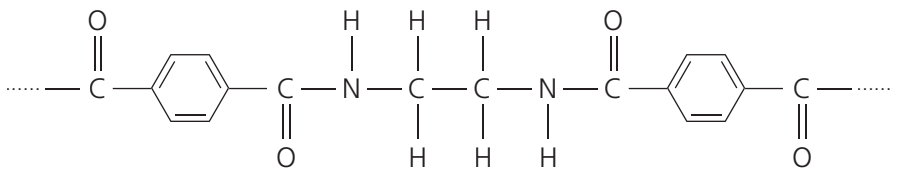


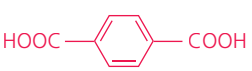
c) There are strong hydrogen bonds between chains of Twaron.

Practice 33.4

1 Nylon is commonly used to make parachutes. Suggest TWO properties of nylon which make it suitable for this application. Strong and lightweight

2 A section of the polymer chain of a polymer is shown below.



a) Draw the structures of the monomers from which the polymer is made.  $H_2NCH_2CH_2NH_2$

b) Write a chemical equation for the production of the polymer.
Refer to the previous page.

Have you mastered?

Key terms

aspirin 阿士匹靈 172 acetylsalicylic acid 乙酰水楊酸 172 paracetamol 撲熱息痛 173

surface tension 表面張力 174 soapy detergent 皂性清潔劑 174

soapless detergent 非皂性清潔劑 174 synthetic detergent 合成清潔劑 174

wetting agent 濕潤劑 175 emulsifying action 乳化作用 175 anionic detergent 陰離子清潔劑 175

hydrophobic 疏水性 175 hydrophilic 親水性 175 sulphonate group 磺酸基 176

emulsion 乳狀物 178 saponification 皂化作用 180 triglyceride 甘油三酯 182

non-biodegradable 生物不可降解的 185 biodegradable 生物可降解的 185 polyester 聚酯 186

poly(ethylene terephthalate) 聚對苯二甲酸乙二酯 187 condensation polymerization 縮合聚合作用 188

nylon 尼龍 191 polyamide 聚酰胺 191

Checklist

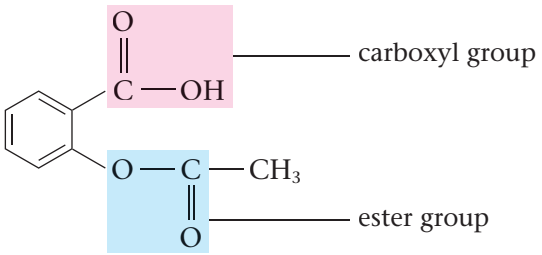
After studying this unit, you should be able to

- ☐ identify the functional groups of aspirin;
- ☐ recognize the medical applications of aspirin;
- ☐ describe the structures of soaps and soapless detergents;
- ☐ recognize that soaps can be made from fats and oils;
- ☐ recognize that soapless detergents can be made from chemicals derived from petroleum;
- ☐ explain the wetting and emulsifying properties of detergents in relation to their structures;
- ☐ relate the cleansing action of soaps and soapless detergents to their structures;
- ☐ describe the structures, properties and uses of polyesters and nylon.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 a) The structure of the active ingredient of aspirin tablets, acetylsalicylic acid, is shown below. It contains two functional groups.



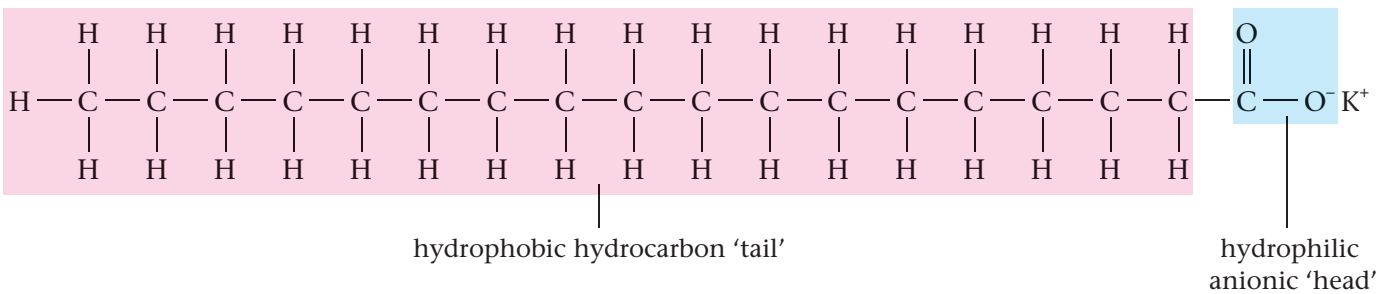
b) Aspirin helps to

- relieve pain;
- relieve fever;
- reduce the risk of heart attack;
- reduce inflammation;
- protect against bowel cancer;
- slow down senile dementia.

2 a) There are two types of detergents:

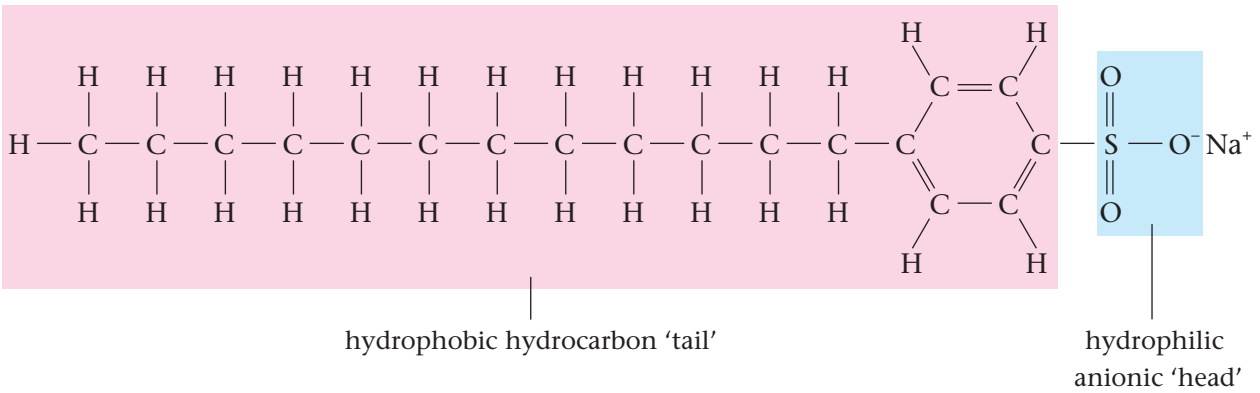
i) soap — made from natural fats and oils;

e.g.



ii) soapless detergents — made from chemicals derived from petroleum.

e.g.



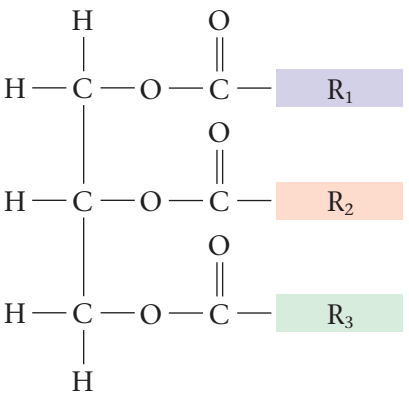
b) A detergent helps water to remove dirt by

i) its ability to act as a wetting agent; and

ii) its emulsifying action.

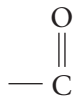
Have you mastered?

3 a) Fats and oil are mixtures of different triglycerides. A triglyceride is formed when glycerol combines with three long-chain carboxylic acids (fatty acids).



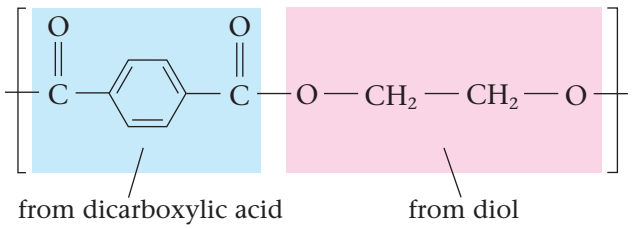
a triglyceride

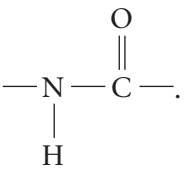
b) Alkaline hydrolysis of triglycerides (i.e. saponification) produces glycerol and a mixture of salts of long-chain carboxylic acids (i.e. soap).

4 a) Polyesters contain the ester linkage  .

b) Polyesters are thermoplastics formed via condensation polymerization.

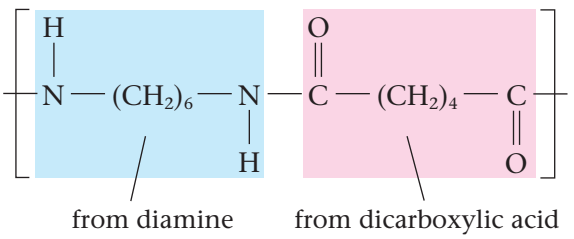
c) The repeating unit of poly(ethylene terephthalate) is



5 a) Nylons are polyamides which contain the amide linkage  .

b) Nylons are thermoplastics formed via condensation polymerization.

c) The repeating unit of nylon-6,6 is



d) Nylons have high melting points and great strength due to the hydrogen bonds between the polymer chains.

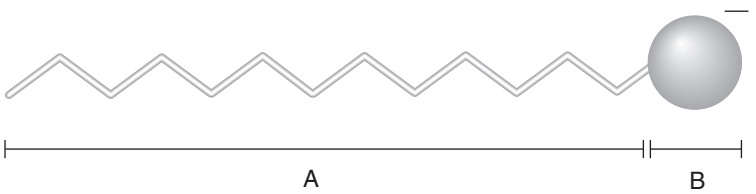
Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 The structure of a typical anionic detergent can be represented as below:



where  represents a hydrocarbon tail

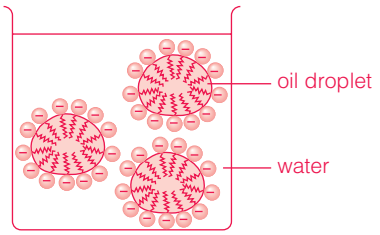
and  represents an ionic part attached to the hydrocarbon.

a) i) Which part is hydrophilic? B

ii) Which part is hydrophobic? A

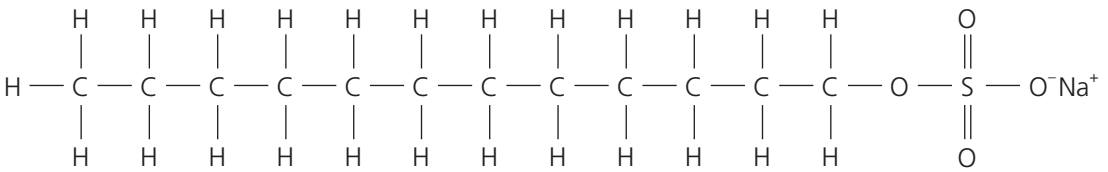
b) Using the above representation, draw a diagram to show how the detergent can suspend an oil drop in water.

b)



2 The structure of a certain detergent is shown below.

Refer to page T63.



The detergent can be used to clean a table cloth stained with grease. Explain the cleansing action of the detergent with reference to its structure.

3 A nylon can be made from the following monomers.

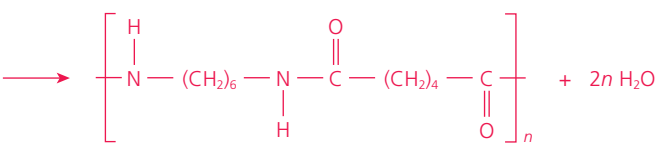
$HOOC(CH_2)_4COOH$ (A) $\qquad$ $H_2N(CH_2)_6NH_2$ (B)

a) Give the systematic names of compounds A and B. A hexanedioic acid; B hexane-1,6-diamine

b) Write a chemical equation for the production of this nylon.

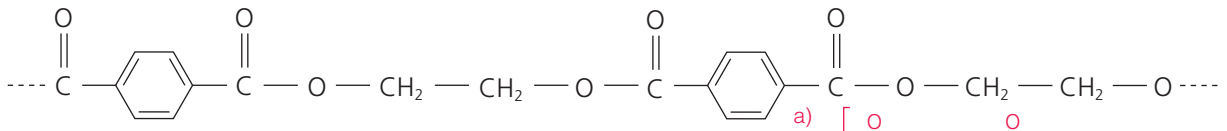
c) Name the type of polymerization involved. Condensation polymerization

b) $n\ HOOC(CH_2)_4COOH + n\ H_2N(CH_2)_6NH_2 \rightarrow$

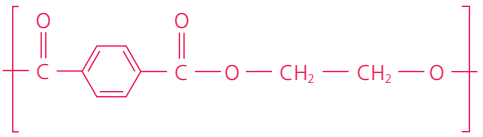


$+ 2n\ H_2O$

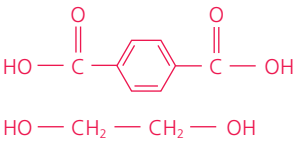
4 A segment of a polyester chain is shown below.



a) Draw the repeating unit of the polyester.

a) 

b) i) Draw the structures of the two monomers used in making the polyester.

b) i) 

ii) Name the type of polymerization involved.
Condensation polymerization

c) Suggest TWO types of material that can be made from the polyester.

c) Fibres used to make clothes
Bottles for carbonated drinks

d) How can the polyester be degraded in the environment?
By hydrolysis in an acidic environment

Part II Multiple choice questions

5 Which of the following statements concerning a soapy detergent is correct?

A It can increase the surface tension of water.
(B) It contains a hydrophobic hydrocarbon chain.
C It can be manufactured from petroleum products.
D It contains a positive ionic part for carrying out emulsification.

(HKCEE, Paper 2, 2008, 35)

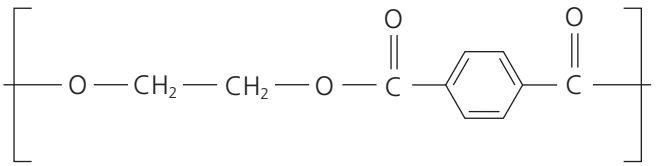
6 In an experiment, a mixture of coconut oil and excess concentrated sodium hydroxide solution is heated for some time. Then a small amount of concentrated sodium chloride solution is added to the reaction mixture with stirring. A solid product is eventually formed. Which of the following statements concerning this experiment is correct?

A The solid formed is glycerol.
B This experiment involves emulsification.
C The purpose of this experiment is to prepare a soapless detergent.
(D) The purpose of adding concentrated sodium chloride solution is to salt out the product formed.

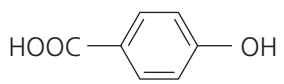
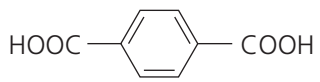
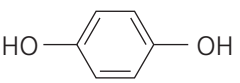
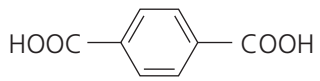
(HKCEE, Paper 2, 2011, 39)

Directions: Questions 7 and 8 refer to the following information:

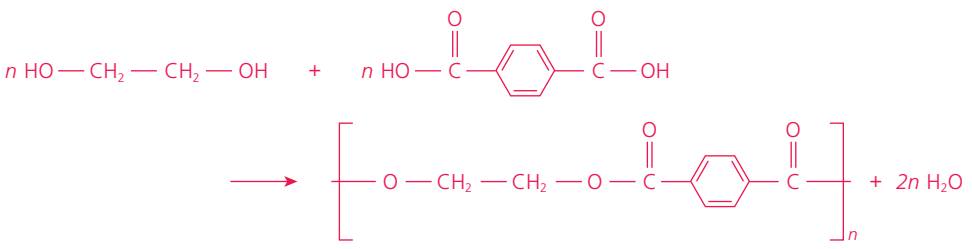
An example of a polyester is



7 The two monomers needed to form this polymer are

	Monomer 1	Monomer 2
A		$HO(CH_2)_2OH$
(B)		$HO(CH_2)_2OH$
C		$HOOC(CH_2)_2COOH$
D		$HOOC(CH_2)_2COOH$

(Edexcel Advanced GCE, Unit 4, Jun. 2012, 6(a))

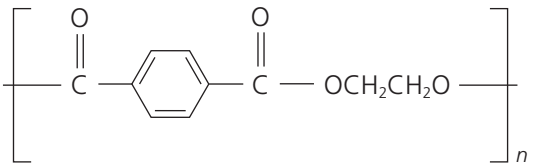
7 

8 The type of reaction to form this polymer is

A addition.
B substitution.
C condensation.
D hydrolysis.

(Edexcel Advanced GCE, Unit 4, Jun. 2012, 6(b))

9 The structure of polymer X is shown below:



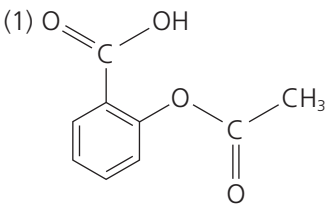
Which of the following statements about X is correct?

A It possesses a ketone functional group.
B It can undergo degradation in an acidic environment.
C It has a giant covalent network structure.
D It has a sharp melting point.

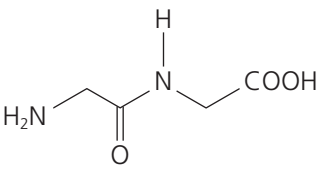
(HKDSE, Practice paper 1A, 2012, 28)

10 Which of the following structures represent(s) the active ingredient(s) in aspirin tablets?

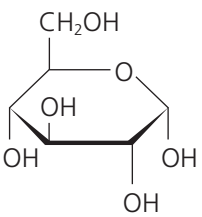
(1)



(2)



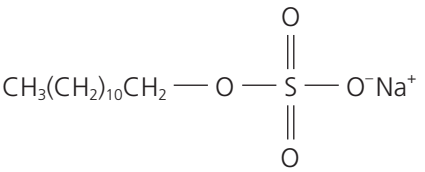
(3)



A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2012, 32)

11 A detergent has the structure shown below.



Which of the following statements concerning this detergent is / are correct?

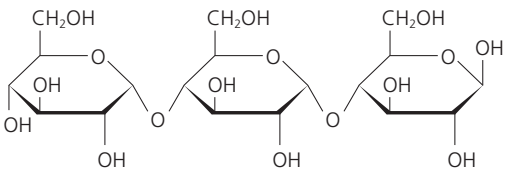
(1) It can be manufactured from vegetable oils.
(2) The part $CH_3(CH_2)_{10}CH_2$— is hydrophobic.
(3) The Na^+ part is responsible for its cleansing action.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

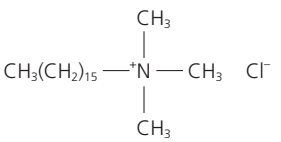
(1) The detergent is a soapless one. It is manufactured from chemicals derived from petroleum.
(3) The Na^+ part is NOT involved in the cleansing process.

12 Consider the following compounds:

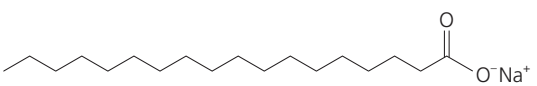
(1)



(2)



(3)



Which of these compounds can be used as active ingredients of detergents?

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Practice paper 1A, 2012, 34)

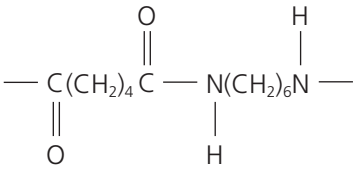
13 Which of the following statements concerning soapy and soapless detergents are correct?

(1) They both are emulsifying agents.
(2) They both contain hydrophobic and hydrophilic parts.
(3) Soapy detergent is biodegradable while soapless detergent is non-biodegradable.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2009, 45)

14 The repeating unit of polymer X is shown below.
Refer to page T63 for explanation.



Which of the following statements about X is / are correct?

(1) X is an addition polymer.
(2) X is formed from two different monomers.
(3) X is used in making packaging materials for electrical appliances.

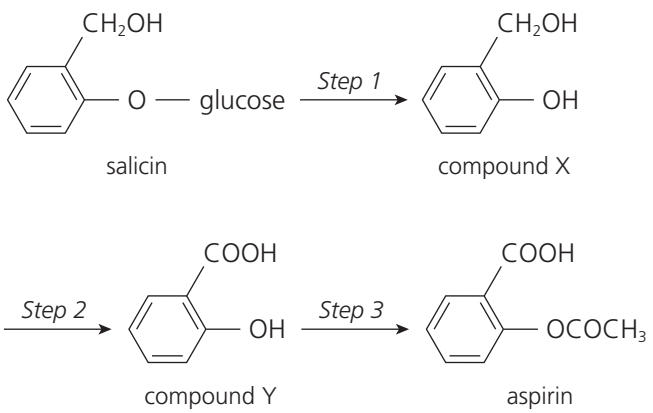
A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

15 Since ancient times, the bark of willow trees has been used for pain relief. In the 19th century, chemists isolated the active compound, salicin, from the bark. This was eventually converted into aspirin, which is now a widely used drug.

The reaction scheme below shows the steps used to carry out the conversion.



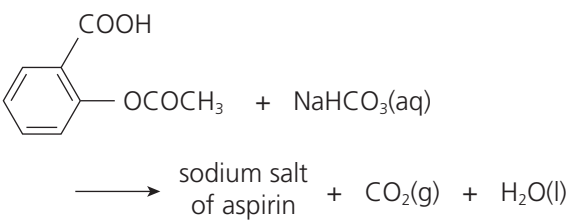
a) *Step 2* involves the conversion of compound X into compound Y.

i) What type of reaction is involved in *Step 2*?
Oxidation

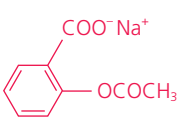
ii) Suggest a suitable reagent to carry out the conversion.
Acidified potassium dichromate solution

b) Suggest a suitable reagent and state the conditions for *Step 3*. Heat ethanoic acid and compound Y in the presence of concentrated sulphuric acid.

c) The sodium salt of aspirin is more soluble than aspirin itself. This salt may be synthesized by the reaction between aspirin and sodium hydrogen carbonate as shown below.



15 c) i)



i) Give the complete structure for the sodium salt of aspirin.

ii) Explain why the sodium salt of aspirin is more water-soluble than aspirin. The ion-dipole interactions between $-COO^-$ and H_2O are stronger than the hydrogen bonds between $-COOH$ and H_2O.

16 The virus H1N1 can cause influenza. It has an oil-based coating.

Washing hands with soapy detergent can help reduce influenza infection caused by the virus H1N1. Suggest why soapy detergent can destroy the virus.

(HKCEE, Paper 1, 2010, 5(a))

17 The procedures in an experiment are summarized below.

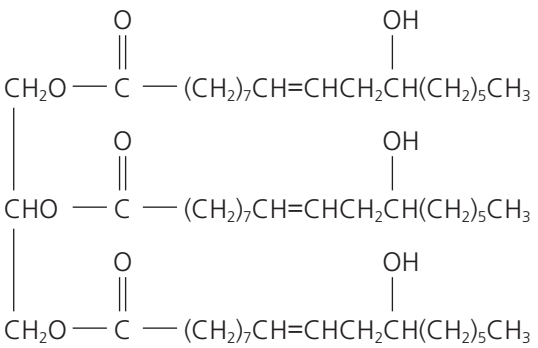
A mixture of castor oil and sodium hydroxide solution was heated gently with stirring for 15 minutes. After cooling down the mixture, a white solid X was obtained upon adding a colourless solution Y. X was then separated out and washed with distilled water.

A small amount of X was put in a test tube containing a mixture of water and a few drops of oil. The contents of the test tube were thoroughly shaken and the observation was recorded.

a) Name the type of reaction involved when the mixture of castor oil and sodium hydroxide solution was heated.

b) Suggest what Y would be.

c) The structure of a main ingredient of castor oil is shown below.



Suggest a structure for X.

d) State the expected observation while shaking the test tube. Explain your answer.

e) Suggest a title for the experiment that reflects its objectives.

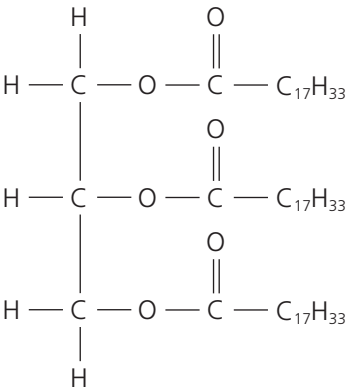
(HKCEE, Paper 1, 2009, 12(a)–(e))

18 One of the ingredients in a washing-up liquid is a detergent.

Write about how the detergent helps to remove fat and grease from dirty plates.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2009, 1(b))

19 An unsaturated fat F is a component of a vegetable oil. The structure of F is shown below:



a) Vegetable oils can be used to make soap.

i) Write the chemical equation involved for the formation of soap from F.

ii) In the presence of an acid, the soap formed in (i) can react with methanol to give compound G, which can be used as a biodiesel. Draw the structure of G.

b) With reference to their relative molecular masses and physical properties, explain why G can be used as a fuel for cars, but F cannot.

(HKDSE, Paper 1B, 2013, 14(b), (c))

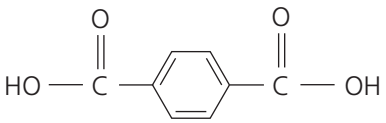
18 When we add a detergent solution to dirty plates, the hydrocarbon 'tails' of the detergent particles dissolve in the grease. The anionic 'heads' of the detergent particles are insoluble in the grease and remain outside.

The surrounding water molecules attract the anionic 'heads' and lift the grease off the plates.

By stirring, the grease breaks up into tiny droplets suspended in the water. These tiny droplets cannot come together again due to the repulsion between the anionic 'heads' of detergent particles on their surfaces. An emulsion forms.

Rinsing washes away the greasy suspension, and leaves the plates clean.

20 There is an increasing demand for PBT, a polymer used in making mouldings for cars. PBT is made using the two monomers shown below.



monomer X

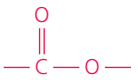


monomer Y

a) Give the systematic name of Y.
butane-1,4-diol

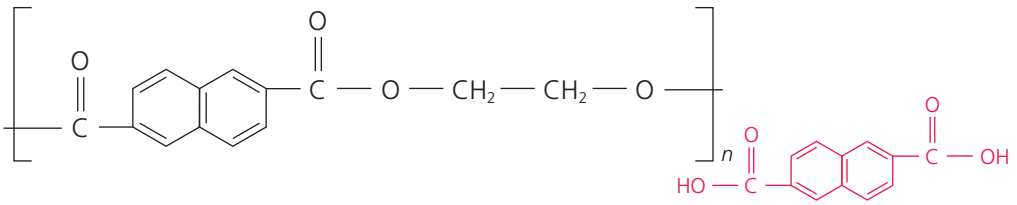
b) Write a chemical equation for the production of PBT.
Refer to page T63.

c) The repeating unit of PBT contains an ester linkage. Write the structural formula of an ester linkage.



d) What characteristic in the structures of X and Y enables them to act as monomers?
Each contains two functional groups.

21 PEN is a polyester made from a carboxylic acid and an alcohol. It is formed from the polymerization of two different monomers and has the following structure:

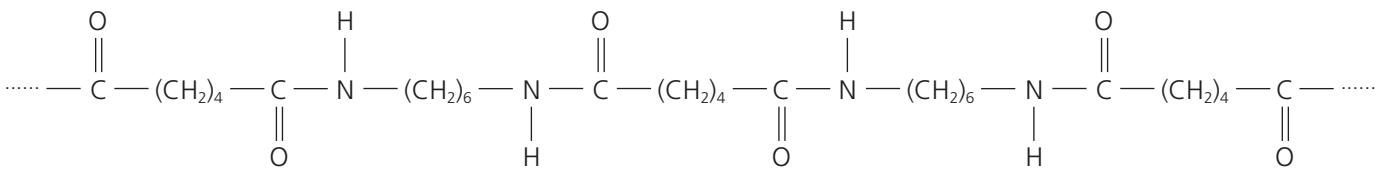


a) Draw the structures of two monomers that can be polymerized to form PEN.

b) Give the systematic name of the alcohol used as a monomer to make PEN.
ethane-1,2-diol

c) Name the type of polymerization involved in the production of PEN.
Condensation polymerization

22 Nylon is a polymer that can be used to make carpets. A portion of the nylon structure is shown below:



a) Draw the structures of two monomers which can be used to make the nylon.
Refer to page T64.

b) Write a chemical equation for the production of the nylon.
Refer to page T64.

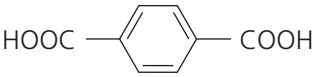
c) Name the functional group connecting the monomer units in nylon.
Amide functional group

d) Nylon is used to replace steel in making gears and bearings. Suggest TWO advantages of using nylon over steel in this application.
Any two of the following:
- More corrosion resistant
- Needs no lubricant
- Strong but light in weight
- Can be moulded to different shapes easily

///23 Kevlar is a polymer made from the following two monomers:



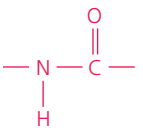
monomer X



monomer Y

a) Write a chemical equation for the production of Kevlar.
Refer to page T64.

b) Draw the structure of the functional group connecting the monomer units in Kevlar.



c) Kevlar is formed by a condensation reaction. Use the reaction between monomers X and Y to explain the meaning of 'condensation'.

During the production of Kevlar, the monomer molecules X and Y react to form polymer molecules, releasing water molecules. A condensation reaction occurs.

A condensation reaction is a chemical reaction in which two or more molecules react together to form a larger molecule with the elimination of a small molecule.

///24 Mandelic acid (2-phenyl-2-hydroxyethanoic acid), $C_6H_5CH(OH)COOH$, is used in some skin creams and can be converted into a condensation polymer.

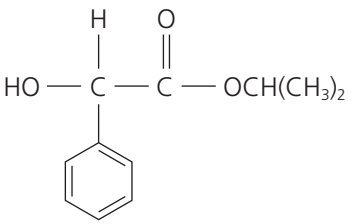
The addition polymer of ethyl methacrylate (ethyl 2-methylprop-2-enoate), $CH_2=C(CH_3)COOC_2H_5$, is used to make some artificial fingernails.
Refer to page T64.

a) Draw two repeating units of a polymer that is formed when

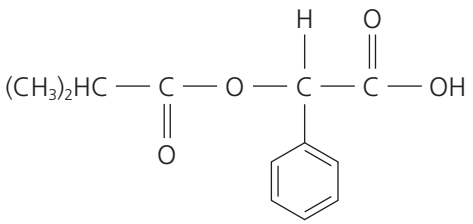
i) mandelic acid, $C_6H_5CH(OH)COOH$, polymerizes;

ii) ethyl methacrylate, $CH_2=C(CH_3)COOC_2H_5$, polymerizes.

b) Mandelic acid has anti-bacterial properties and is used in some skin creams. A cosmetic chemist used mandelic acid to prepare two different esters that might be suitable for new skin creams. The structures of the two esters are shown below.



ester 1



ester 2

i) Draw the structure of a carbon compound that could react with mandelic acid, $C_6H_5CH(OH)COOH$, to produce ester 1.

ii) Identify a carbon compound that could react with mandelic acid to produce ester 2.

iii) Ester 1 is less soluble in water than mandelic acid, $C_6H_5CH(OH)COOH$.

Explain the difference in water solubility of mandelic acid and ester 1. You may use a labelled diagram in your answer.

iv) Before any skin cream can be sold to the public, it must be tested to ensure it is safe to use.

Suggest why.

(OCR Advanced GCE, Chem. A, F324, Jun. 2011, 3(b), (d))

Unit Exercise

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Compound X has the following structure:

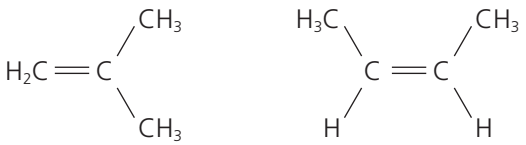
$CH_2=CHCH_2OH$

The systematic name of X is

A prop-1-en-3-ol.
(B) prop-2-en-1-ol.
C 3-hydroxypropene.
D 1-hydroxyprop-3-ene.

(HKDSE, Paper 1A, 2012, 11)

!!2 Consider the compounds X and Y shown below:



X

Y

Which of the following statements about X and Y is correct?

A X and Y are a pair of *cis-trans* isomers.
(B) Both X and Y react with $H_2(g)$ in the presence of Ni(s).
C X and Y react separately with Br_2 in CH_3CCl_3 to give the same organic product.
D Both the polymerization of X and that of Y give the same addition polymer.

(HKDSE, Paper 1A, 2013, 31)

4 $CH_3CHBrCH=CHCl$ contains one chiral carbon. It has enantiomers.

chiral carbon

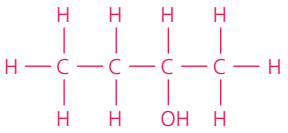
$CH_3CHBrCH=CHCl$

$CH_3CHBrCH=CHCl$ has two different atoms or groups of atoms on each carbon atom of the carbon-carbon double bond. It has *cis*-trans isomers.

5 $CH_3CH=CH_2(g) + [O] + H_2O(l) \longrightarrow CH_3CH(OH)CH_2OH(aq)$

[O] from acidified potassium permanganate solution

3 Butan-2-ol has two alkyl groups attached to the carbon atom bearing the hydroxyl group. Thus, it is a secondary alcohol.



butan-2-ol

3 Which of the following is a secondary alcohol?

A Butan-1-ol
(B) Butan-2-ol
C Methylpropan-1-ol
D Methylpropan-2-ol

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 13)

!!4 Which of the following compounds has both enantiomers and *cis-trans* isomers?

A $CH_3CH=CHCH_2CH_3$
B $CH_3CHClCH=C(CH_3)_2$
C $CH_3CCl=CClCH_3$
(D) $CH_3CHBrCH=CHCl$

(Edexcel Advanced GCE, Unit 4, Jun. 2010, 4)

5 If propene, $CH_3CH=CH_2$, is reacted with acidified potassium permanganate solution the organic product is

(A) $CH_3CH(OH)CH_2OH$.
B $CH_3CH(OH)CH_3$.
C $HOCH_2CH_2CH_2OH$.
D $CH_3CH_2CH_2OH$.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 17)

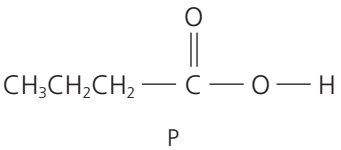
6 Which of the following liquids would react with phosphorus pentachloride to give a gas which fumes in moist air?

A Ethanal
(B) Ethanol
C Tetrachloromethane
D Trichloromethane

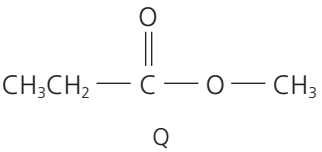
(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2011, 7(b))

$CH_3CH_2OH(l) + PCl_5(s) \longrightarrow CH_3CH_2Cl(l) + POCl_3(l) + HCl(g)$

7 The structures of compounds P and Q are shown below:



P



Q

Which of the following statements is correct?

A P and Q are both acids.
B P is more volatile than Q.
(C) P dissolves in water readily but Q does not.
D P and Q both decolourize aqueous bromine rapidly.

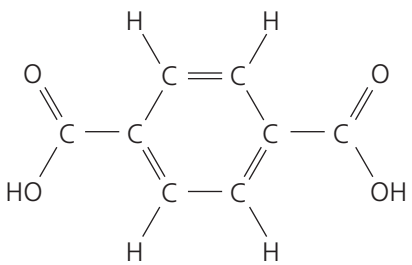
(HKCEE, Paper 2, 2010, 18)

8 Which of the following statements concerning compound U ($CH_3CH_2CH{=}CHCH_2CH_2OH$) is correct?

A The empirical formula of U is C_3H_6O.
B The systematic name of U is hex-4-en-1-ol.
C U reacts with HCl to give a single product.
(D) U can separately turn Br_2(aq) and acidified $KMnO_4$(aq) colourless.

(HKDSE, Paper 1A, 2012, 29)

9 The two compounds shown below undergo condensation polymerization under suitable conditions.



and

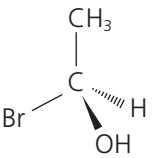


What small molecule would be eliminated in this condensation polymerization?

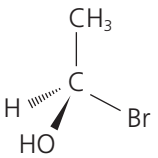
(A) H_2O
B H_2O_2
C CH_3OH
D HCOOH

(HKCEE, Paper 2, 2011, 34)

10 The three-dimensional structures of a molecule of compound X and that of compound Y are shown below:



X



Y

Which of the following statements about X and Y is / are correct?

(1) X and Y are a pair of structural isomers.
(2) X and Y have the same boiling point.
(3) X is optically active while Y is not.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1) X and Y are a pair of enantiomers.
(3) Both X and Y are optically active.

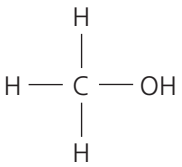
11 Which of the following compounds can be formed when $(CH_3)_2C(OH)CH_2CH_3$ is dehydrated?

(1) $(CH_3)_2C{=}CHCH_3$
(2) $(CH_3)_2CHCH{=}CH_2$
(3) $CH_2{=}C(CH_3)CH_2CH_3$

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2012, 33)

12 An organic compound has the following structure:



Which of the following statements about this compound is / are correct?

(1) It is immiscible with water.
(2) It is neutral to litmus solution.
(3) It burns with a non-luminous flame.

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Paper 1A, 2013, 20)

13 Which of the following compounds can be oxidized by acidified $Na_2Cr_2O_7(aq)$ at room temperature?

(1) CH_3CH_2OH
(2) CH_3COCH_3
(3) $(CH_3)_3COH$

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKASLE, Paper 1, 2008, 5(d))

Directions : *Each question (Questions 14–16) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

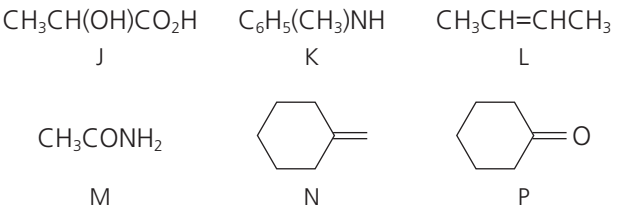
A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
14	Propane can change acidified potassium permanganate solution from purple to colourless.	Substitution reaction occurs when propane is added to acidified potassium permanganate solution *(HKCEE, Paper 2, 2011, 50)*	D
15	2-chlorobut-1-ene shows *cis-trans* isomerism.	2-chlorobut-1-ene has a double bond. *(HKDSE, Paper 1A, 2012, 36)*	C
16	Discharge of synthetic detergents into rivers may cause rapid growth of algae.	Synthetic detergents may contain nutrients for the growth of algae. *(HKCEE, Paper 2, 2009, 50)*	A

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

17 Consider the structural formulae, J, K, L, M, N and P, shown below:



For each of the following descriptions, choose one formula from above that matches the description.

a) A compound which is optically active

b) A compound which forms a condensation polymer under suitable conditions

c) A compound which exhibits *cis-trans* isomerism

(HKASLE, Paper 1, 2011, 7(a), (c), (d))

18 In this question, you are asked to suggest structures for several carbon compounds.

Compounds F, G and H are unbranched alkenes that are isomers, each with a relative molecular mass of 70.0.

Compounds F and G are *cis-trans* stereoisomers.

Compound H is a structural isomer of compounds F and G.

a) Explain what is meant by the terms 'structural isomer' and 'stereoisomer'.

b) Explain why some alkenes have *cis-trans* isomerism.

c) Analyze this information to suggest possible structures for compounds F, G and H.

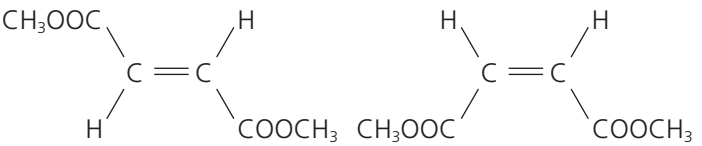
Refer to page T66.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2009, 7(a))

18 a) Structural isomer — compounds with the same molecular formula but differs in the order in which atoms are linked
Stereoisomers — compounds with their atoms linked in the same way but differs in the spatial arrangement of their atoms

b) *Cis-trans* isomerism occurs in some alkenes because
- rotation about the carbon-carbon double bond is restricted;
- there are different atoms or groups of atoms on each carbon atom of the carbon-carbon double bond.

19 Dimethyl fumarate can be found in most leather products since it is commonly used as a mould inhibitor. However, it was banned in Europe for all kinds of consumer goods in March 2009 because it was found to cause skin allergies. Compound A is an isomer of dimethyl fumarate. The structures and melting points of these two compounds are given below:



a) Name the type of isomerism involved.

b) Explain why the melting point of A is lower than that of dimethyl fumarate.

(HKASLE, Paper 1, 2010, 6(a))

20 A is an alkanol with three carbon atoms and one oxygen atom in its molecule. A reacts with acidified potassium dichromate solution to form compound B. In the presence of a small amount of concentrated sulphuric acid, A reacts with B to form compound C. C can be separated from the reaction mixture and has a pleasant smell.

a) Write the structural formulae of A, B and C.

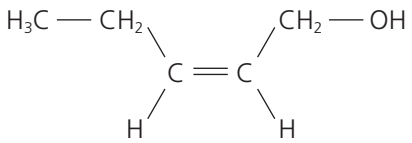
b) State the expected observation for the reaction of A with acidified potassium dichromate solution.

c) Suggest a method to separate C from the reaction mixture.

d) A compound has the same molecular formula as A but a different structure from A. Suggest a structural formula for this compound.

(HKCEE, Paper 1, 2010, 3)

21 'Violet oil' is sometimes used in aromatherapy treatments for its mild pain-killing properties. The oil has a strong 'leafy' odour due partly to the presence of compound A.



compound A

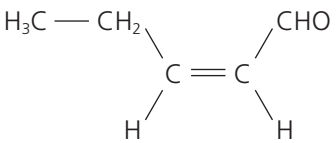
a) Compound A is one of a pair of *cis-trans* isomers.

i) Draw a diagram to show the structure of the other isomer.

ii) Explain why $CH_3CH_2CH{=}CHCH_2OH$ can exist as a pair of *cis-trans* isomers.

b) Compound A reacts with hydrogen chloride to form TWO products with the molecular formula $C_5H_{10}Cl_2$. Draw the structures of both of these products.
Refer to page T66.

c) A student produces compound B by gently heating compound A with suitable reagents and immediately distilling the product from the mixture.



compound B

i) Name the reagents the student uses to convert compound A into compound B.
Acidified potassium dichromate solution

ii) Name the functional group that is present in compound B but NOT in compound A.
Carbonyl group

d) A student carries out a reaction using the same reagents as in part (c), but by heating the reaction mixture under reflux. Compound C is produced.

i) Explain what is meant by the term 'heating under reflux'.
Boil a liquid with a vertical condenser.

ii) Draw the structure of the new functional group present in compound C and name this functional group.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2012, 2(d), (e), (g), (h))

22 Four isomers, each with the molecular formula $C_4H_{10}O$, are shown below.

Isomer A: $CH_3CH_2CH_2CH_2OH$

Isomer B: $CH_3CH_2CH(OH)CH_3$

Isomer C: $(CH_3)_3COH$

Isomer D: $CH_3CH(CH_3)CH_2OH$

a) Which isomer is a secondary alcohol? Justify your answer.
Isomer B
There are two alkyl groups attached to the carbon bearing the hydroxyl group.

b) Which isomer is resistant to oxidation when heated with acidified potassium dichromate solution? Justify your answer in terms of the structure of the isomer.

c) Which isomer can be oxidized to a ketone? Draw the structure of the ketone produced.

d) Which isomers can be oxidized to an aldehyde?
Isomers A and D

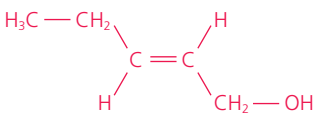
e) Phosphorus pentachloride, PCl_5, is used to test for the presence of an –OH group.

What would you expect to see when any of the above four isomers, A, B, C or D, are reacted with phosphorus pentachloride?
Steamy / misty / white fumes

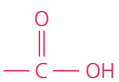
f) Complete the equation for the reaction shown below. State symbols are NOT required.

$$C_4H_9OH + PCl_5 \longrightarrow C_4H_9Cl + POCl_3 + HCl$$

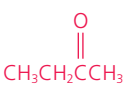
(Edexcel Advanced Subsidiary GCE, Unit 2, Jun. 2010, 20(b))

21 a) i) 

ii) Rotation about the carbon-carbon double bond is restricted.
There are different atoms or groups of atoms on each carbon atom of the carbon-carbon double bond.

d) ii) 
carboxyl group

22 b) Isomer C
because it is a tertiary alcohol / there is no hydrogen atom to be removed from the carbon bearing the hydroxyl group.

c) Isomer B


23 Hydroxyethanal, $HOCH_2CHO$, is sometimes referred to as the 'first sugar' as it is the simplest possible molecule that contains both an aldehyde group and an alcohol group.

A biochemist investigated some redox reactions of hydroxyethanal and found that several different products were produced.

a) The biochemist reacted hydroxyethanal with acidified dichromate solution by heating under reflux.

Write an equation for this oxidation. Use [O] to represent the oxidizing agent.

$HOCH_2CHO + 3[O] \longrightarrow HOOCCOOH + H_2O$

b) The biochemist then reduced hydroxyethanal using aqueous $NaBH_4$.

Write the structural formula of the organic product. $HOCH_2CH_2OH$

(OCR Advanced GCE, Chem. A, F324, Jan. 2010, 2(b)–(c)(i))

24 Consider the conversions of organic compounds shown below:

$$\underset{\text{X}}{CH_3CH_2COCH_2CH_3} \xrightarrow{R} \underset{\text{Y}}{CH_3CH_2CH(OH)CH_2CH_3}$$

$$\xrightarrow[\text{agent}]{\text{dehydrating}} \underset{\text{Z}}{\text{a mixture of alkenes}}$$

a) Suggest a chemical test to distinguish between X and Y.

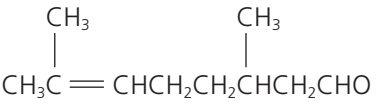
b) Suggest what reagent R might be.

c) The mixture Z contains two alkenes with the same structural formula. Draw the respective structures of these two alkenes, and state their isomeric relationship.

d) The alkenes in (c) can react with HCl to form an optically active chloroalkane. Write the structural formula of this chloroalkane.

(HKDSE, Paper 1B, 2013, 15)

25 A naturally occurring organic compound has the following structure:



a) Suggest a systematic name for this compound.

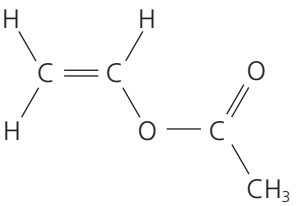
b) Give the structure of the major organic product(s) formed when this compound reacts separately with the following reagents:

i) HCl

ii) $LiAlH_4$ and then H_3O^+

(HKALE, Paper 2, 2012, 13(a), (b)(i)–(ii))

26 Poly(ethenyl ethanoate) is a polymer. Its monomer is ethenyl ethanoate with the structure shown below:



a) Ethene is the raw material used in making ethenyl ethanoate. Ethene can be produced from hydrocarbons of higher molecular mass by an important industrial process.

i) Name this industrial process.

ii) Explain why this process is important.

b) Draw the structure of poly(ethenyl ethanoate).

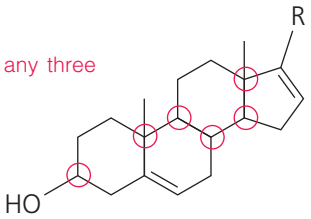
c) Ethyl ethanoate is an organic solvent.

i) Draw the structure of ethyl ethanoate.

ii) Suggest a chemical test to show how to distinguish between ethenyl ethanoate and ethyl ethanoate.

(HKDSE, Paper 1B, 2012, 2)

27 Abiraterone, $C_{24}H_{31}NO$, is a new drug under trial as a treatment for cancer of the prostate gland. The structure of abiraterone is shown below.



any three

R is a ring structure with the molecular formula C_5H_4N

abiraterone

a) Name TWO functional groups present in the part of the abiraterone molecule shown.
Carbon-carbon double bond
Hydroxyl group

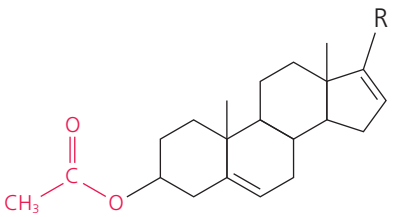
b) A molecule of abiraterone is chiral because it has a number of chiral carbon atoms.

i) On the structural formula shown above, circle three of these chiral carbon atoms.

ii) A molecule with an asymmetric carbon atom exists as two enantiomers. Explain why these are different compounds. Enantiomers are mirror images that are not superposable.

c) In treating cancers, abiraterone is used as its ethanoate ester.

i) On the diagram below, draw the full structure of the ethanoate group in the correct position.



ii) Name the TWO substances that must be heated under reflux with abiraterone to make its ethanoate ester.
Ethanoic acid, concentrated sulphuric acid

d) Complete the equation below for the reaction of abiraterone with excess bromine. Assume that the 'R group' in the structural formula of abiraterone does not react with bromine.

$C_{24}H_{31}NO + 2Br_2 \longrightarrow C_{24}H_{31}NOBr_4$

e) Abiraterone ethanoate is a white solid which can be purified by re-crystallization. Re-crystallization removes both soluble and insoluble impurities.

Describe how a student would carry out this procedure using a suitable solvent.

f) Clinical trials are carried out to determine the suitability of abiraterone as a drug.

Give three questions that clinical trials are designed to answer.

(OCR Advanced GCE, Chem. B (Salters), F334, Jun. 2010, 1(a)–(b), (d)–(f))

27 e) Use the minimum amount of hot solvent to dissolve the solid.
Filter to remove the insoluble impurities.
Allow the filtrate to cool slowly for crystals to re-form.
Filter off the crystals.
Wash and dry.

f) Is the drug safe?
Does the drug do the job it is designed to do?
Is it better than the standard treatment being used?

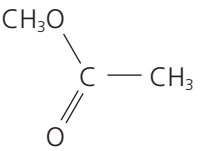
28 The following compounds can be distinguished by observing what happens in test tube reactions.

For each pair, suggest a suitable reagent or reagents that could be added separately to each compound in order to distinguish them.

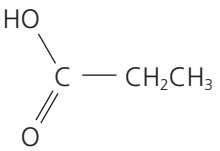
Describe what you would observe with each compound.

Refer to page T66.

a)

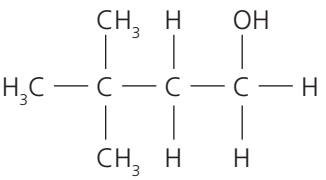


E

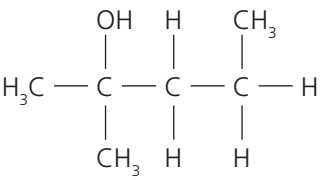


F

b)



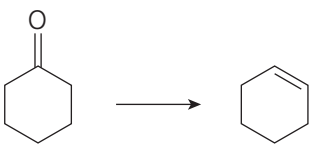
J



K

(AQA Advanced GCE, Unit 4, Jun. 2011, 7(a)(i), (iii))

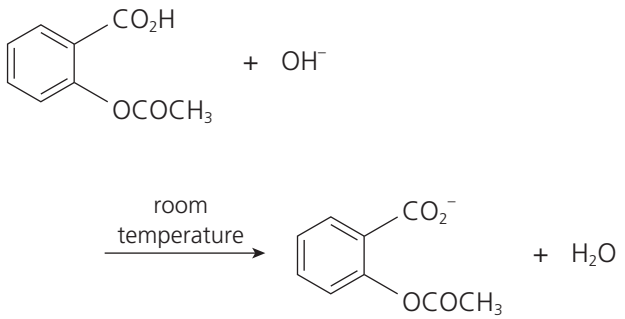
29 Outline a synthetic route, with no more than three steps, to accomplish the following conversion. For each step, give the reagent(s), reaction conditions and structure of the organic product.



(HKDSE, Practice paper 1B, 2012, 11)

Topic Exercise

30 A commercial aspirin sample E was known to contain about 90% by mass of aspirin, while the rest was an inert binder. Based on the following reaction, a student designed an experiment and performed it at room temperature to determine the percentage by mass of aspirin in E.



The student added 2.25 g of E to 25.00 cm^3 of 3.05 mol dm^{-3} NaOH(aq), and then back titrated the excess NaOH(aq) with 2.50 mol dm^{-3} HCl(aq). The volume of HCl(aq) used was 23.10 cm^3.

a) Suggest an indicator for the titration.

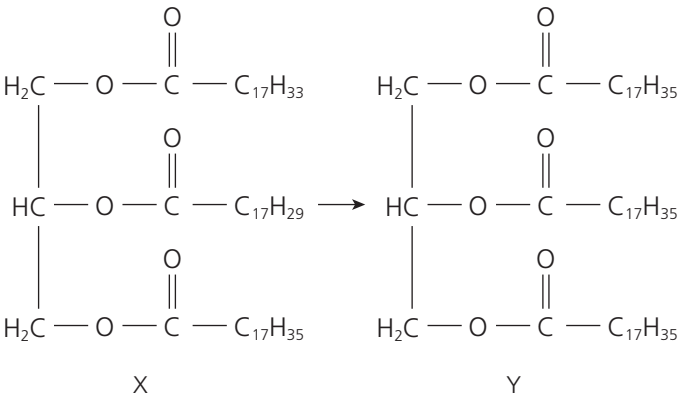
b) From the student's experimental results, calculate the percentage by mass of aspirin in E. Suggest why the calculated percentage by mass of aspirin deviates greatly from 90%.

(Relative molecular mass of aspirin = 180.0)

c) Suggest ONE improvement to the design of the experiment to find the percentage by mass of aspirin in E.

(HKASLE, Paper 1, 2012, 10)

31 The diagram below shows the conversion of an oil molecule X to a fat molecule Y.



a) i) Given that all alkyl groups in both X and Y are straight chains, label the chiral carbon(s) by using '*' in the above diagram.

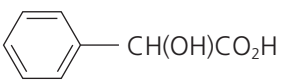
ii) With reference to (i), explain whether a change in optical activity is involved in the above conversion.

b) One of the products in the alkaline hydrolysis of Y has a cleansing property. Explain the cleansing property of this product.

(Communication mark will be awarded)

(HKDSE, Paper 1B, 2012, 14)

32 The structural formula shown below can represent two compounds with the same melting point and same solubility in water.



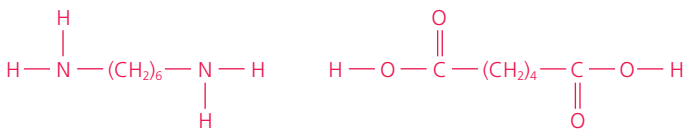
a) i) Draw a three-dimensional structure for each of the two compounds.

ii) State ONE difference in physical properties of these compounds.

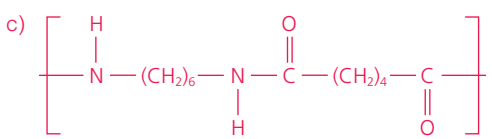
b) Both compounds can undergo polymerization under suitable conditions. Draw the repeating unit of the polymer formed from one of these compounds.

(HKDSE, Practice paper 1B, 2012, 12)

35 b)

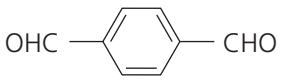


c)

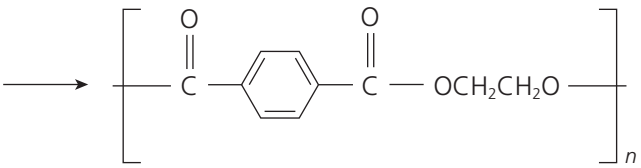


33 Polyethylene terephthalate (PET) is a polymer commonly used in making soft drink bottles.

a) Suggest a synthetic route with *no more than three steps* to convert benzene-1,4-dicarbaldehyde to PET. (You are NOT allowed to use organic reagents with more than three carbon atoms.)



benzene-1,4-dicarbaldehyde

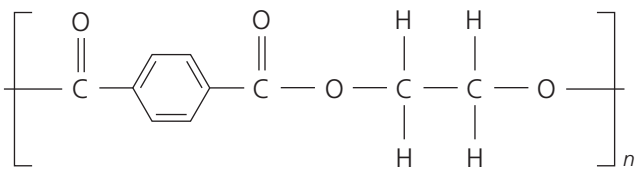


PET

b) Suggest why PET, as compared with polystyrene, is more readily degraded in the environment.

(HKASLE, Paper 2, 2010, 3)

34 Polyethylene terephthalate (PET) is a thermoplastic that can be used to make soft drink bottles. PET is formed from polymerization of two different monomers and has the following structure:



a) Write the structural formulae of two monomers that can be polymerized to PET.

b) Name the type of the above polymerization and give the reason.

c) PET bottles CANNOT be used to store drain cleansers containing concentrated sodium hydroxide solution because PET reacts with strong alkalis.

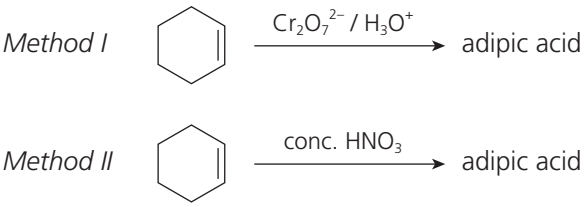
i) What is the reaction involved?

ii) Write the structural formula of one of the products of the reaction.

d) Polypropene (PP) can be used to make bottles for storing drain cleansers containing strong alkalis. Write a chemical equation for the polymerization to form PP.

(HKCEE, Paper 1, 2010, 12(a), (c), (e), (f))

///35 A common type of nylon is called nylon-6,6. It is made from two monomers, one of which is hexanedioic acid, also known as adipic acid. The acid is commonly synthesized from cyclohexene using *Method I* or *Method II* outlined below:



However, both methods are considered as environmentally unfriendly.

a) Name the type of polymerization involved in the production of nylons.
Condensation polymerization

b) Draw the structures of the monomers used to make nylon-6,6.
Refer to the previous page.

c) Draw the repeating unit of nylon-6,6.
Refer to the previous page.

d) Explain, in terms of the types of intermolecular attractions between the polymer chains, why nylon is stronger than polythene.

e) Suppose you drop some dilute sulphuric acid on a fabric made from nylon. What would you expect to observe? Explain your answer.

f) Explain why both *Methods I* and *II*, for the synthesis of adipic acid are considered as environmentally unfriendly.

d) There are hydrogen bonds between the polymer chains of nylon.
Only weak van der Waals' forces exist between the polymer chains of polythene.
Hence nylon is stronger than polythene.

e) A hole would appear because the acid hydrolyzes the amide linkages in nylon.

f) Method I:
The waste contains toxic dichromate ions.
Method II:
Concentrated nitric acid is a strong acid. Discharge of the waste into waterways leads to environmental pollution.